Variable Selection R Codes Examples

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Manual Backward Variable Selection

How backward elimination works
The process follows a clear, ordered loop:
- Fit the full model with all predictors.
- Identify the predictor with the largest p-value.
- If that p-value is > 0.05, remove that predictor. - Refit the model without it.
- Repeat until all remaining predictors have p < 0.05.
This produces a final model where every included variable is individually significant conditional on the others.

```
library(olsrr)
```

```
##
## Attaching package: 'olsrr'

## The following object is masked from 'package:datasets':
##
##     rivers
```

```
library(datasets)

f <- lm(mpg ~., data = mtcars)
summary(f)
```

```
##
## Call:
## lm(formula = mpg ~ ., data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.4506 -1.6044 -0.1196  1.2193  4.6271
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 12.30337   18.71788   0.657   0.5181
## cyl         -0.11144    1.04502  -0.107   0.9161
## disp         0.01334    0.01786   0.747   0.4635
## hp          -0.02148    0.02177  -0.987   0.3350
## drat         0.78711    1.63537   0.481   0.6353
## wt          -3.71530    1.89441  -1.961   0.0633 .
## qsec         0.82104    0.73084   1.123   0.2739
## vs           0.31776    2.10451   0.151   0.8814
## am           2.52023    2.05665   1.225   0.2340
## gear         0.65541    1.49326   0.439   0.6652
## carb        -0.19942    0.82875  -0.241   0.8122
```

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.65 on 21 degrees of freedom
## Multiple R-squared:  0.869,  Adjusted R-squared:  0.8066
## F-statistic: 13.93 on 10 and 21 DF,  p-value: 3.793e-07
```

```r
#cyl has the highest p value of 0.9161
f1<-update(f,.~.- cyl)
summary(f1)
```

```
##
## Call:
## lm(formula = mpg ~ disp + hp + drat + wt + qsec + vs + am + gear +
##     carb, data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.4286 -1.5908 -0.0412  1.2120  4.5961
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 10.96007   13.53030   0.810   0.4266
## disp         0.01283    0.01682   0.763   0.4538
## hp          -0.02191    0.02091  -1.048   0.3062
## drat         0.83520    1.53625   0.544   0.5921
## wt          -3.69251    1.83954  -2.007   0.0572 .
## qsec         0.84244    0.68678   1.227   0.2329
## vs           0.38975    1.94800   0.200   0.8433
## am           2.57743    1.94035   1.328   0.1977
## gear         0.71155    1.36562   0.521   0.6075
## carb        -0.21958    0.78856  -0.278   0.7833
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.59 on 22 degrees of freedom
## Multiple R-squared:  0.8689, Adjusted R-squared:  0.8153
## F-statistic: 16.21 on 9 and 22 DF,  p-value: 9.031e-08
```

```r
#vs has the highest p value of 0.8433
f1<-update(f1,.~.- vs)
summary(f1)
```

```
##
## Call:
## lm(formula = mpg ~ disp + hp + drat + wt + qsec + am + gear +
##     carb, data = mtcars)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -3.356 -1.576 -0.149  1.218  4.604
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  9.76828   11.89230   0.821   0.4199
## disp         0.01214    0.01612   0.753   0.4590
```

```
## hp          -0.02095    0.01993  -1.051   0.3040
## drat         0.87510    1.49113   0.587   0.5630
## wt          -3.71151    1.79834  -2.064   0.0505 .
## qsec         0.91083    0.58312   1.562   0.1319
## am           2.52390    1.88128   1.342   0.1928
## gear         0.75984    1.31577   0.577   0.5692
## carb        -0.24796    0.75933  -0.327   0.7470
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.535 on 23 degrees of freedom
## Multiple R-squared:  0.8687, Adjusted R-squared:  0.823
## F-statistic: 19.02 on 8 and 23 DF,  p-value: 2.008e-08
```

```
#carb has the highest p value of  0.7470
f1<-update(f1,.~.- carb)
summary(f1)
```

```
##
## Call:
## lm(formula = mpg ~ disp + hp + drat + wt + qsec + am + gear,
##     data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.1200 -1.7753 -0.1446  1.0903  4.7172
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  9.19763   11.54220   0.797  0.43334
## disp         0.01552    0.01214   1.278  0.21342
## hp          -0.02471    0.01596  -1.548  0.13476
## drat         0.81023    1.45007   0.559  0.58151
## wt          -4.13065    1.23593  -3.342  0.00272 **
## qsec         1.00979    0.48883   2.066  0.04981 *
## am           2.58980    1.83528   1.411  0.17104
## gear         0.60644    1.20596   0.503  0.61964
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.488 on 24 degrees of freedom
## Multiple R-squared:  0.8681, Adjusted R-squared:  0.8296
## F-statistic: 22.56 on 7 and 24 DF,  p-value: 4.218e-09
```

```
#gear has the highest p value of  0.61964
f1<-update(f1,.~.- gear)
summary(f1)
```

```
##
## Call:
## lm(formula = mpg ~ disp + hp + drat + wt + qsec + am, data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.2669 -1.6148 -0.2585  1.1220  4.5564
##
```

```
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 10.71062   10.97539   0.976  0.33848
## disp         0.01310    0.01098   1.193  0.24405
## hp          -0.02180    0.01465  -1.488  0.14938
## drat         1.02065    1.36748   0.746  0.46240
## wt          -4.04454    1.20558  -3.355  0.00254 **
## qsec         0.99073    0.48002   2.064  0.04955 *
## am           2.98469    1.63382   1.827  0.07969 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.45 on 25 degrees of freedom
## Multiple R-squared:  0.8667, Adjusted R-squared:  0.8347
## F-statistic: 27.09 on 6 and 25 DF,  p-value: 8.637e-10
```

```r
#drat has the highest p value of  0.46240
f1<-update(f1,.~.- drat)
summary(f1)
```

```
##
## Call:
## lm(formula = mpg ~ disp + hp + wt + qsec + am, data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.5399 -1.7398 -0.3196  1.1676  4.5534
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 14.36190    9.74079   1.474  0.15238
## disp         0.01124    0.01060   1.060  0.29897
## hp          -0.02117    0.01450  -1.460  0.15639
## wt          -4.08433    1.19410  -3.420  0.00208 **
## qsec         1.00690    0.47543   2.118  0.04391 *
## am           3.47045    1.48578   2.336  0.02749 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.429 on 26 degrees of freedom
## Multiple R-squared:  0.8637, Adjusted R-squared:  0.8375
## F-statistic: 32.96 on 5 and 26 DF,  p-value: 1.844e-10
```

```r
#disp has the highest p value of  0.29897
f1<-update(f1,.~.- disp)
summary(f1)
```

```
##
## Call:
## lm(formula = mpg ~ hp + wt + qsec + am, data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.4975 -1.5902 -0.1122  1.1795  4.5404
##
## Coefficients:
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 17.44019    9.31887   1.871  0.07215 .
## hp          -0.01765    0.01415  -1.247  0.22309
## wt          -3.23810    0.88990  -3.639  0.00114 **
## qsec         0.81060    0.43887   1.847  0.07573 .
## am           2.92550    1.39715   2.094  0.04579 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.435 on 27 degrees of freedom
## Multiple R-squared:  0.8579, Adjusted R-squared:  0.8368
## F-statistic: 40.74 on 4 and 27 DF,  p-value: 4.589e-11
```

```
#hp has the highest p value of  0.22309
f1<-update(f1,.~.- hp)
summary(f1)
```

```
##
## Call:
## lm(formula = mpg ~ wt + qsec + am, data = mtcars)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.4811 -1.5555 -0.7257  1.4110  4.6610
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   9.6178     6.9596   1.382 0.177915
## wt           -3.9165     0.7112  -5.507 6.95e-06 ***
## qsec          1.2259     0.2887   4.247 0.000216 ***
## am            2.9358     1.4109   2.081 0.046716 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.459 on 28 degrees of freedom
## Multiple R-squared:  0.8497, Adjusted R-squared:  0.8336
## F-statistic: 52.75 on 3 and 28 DF,  p-value: 1.21e-11
```

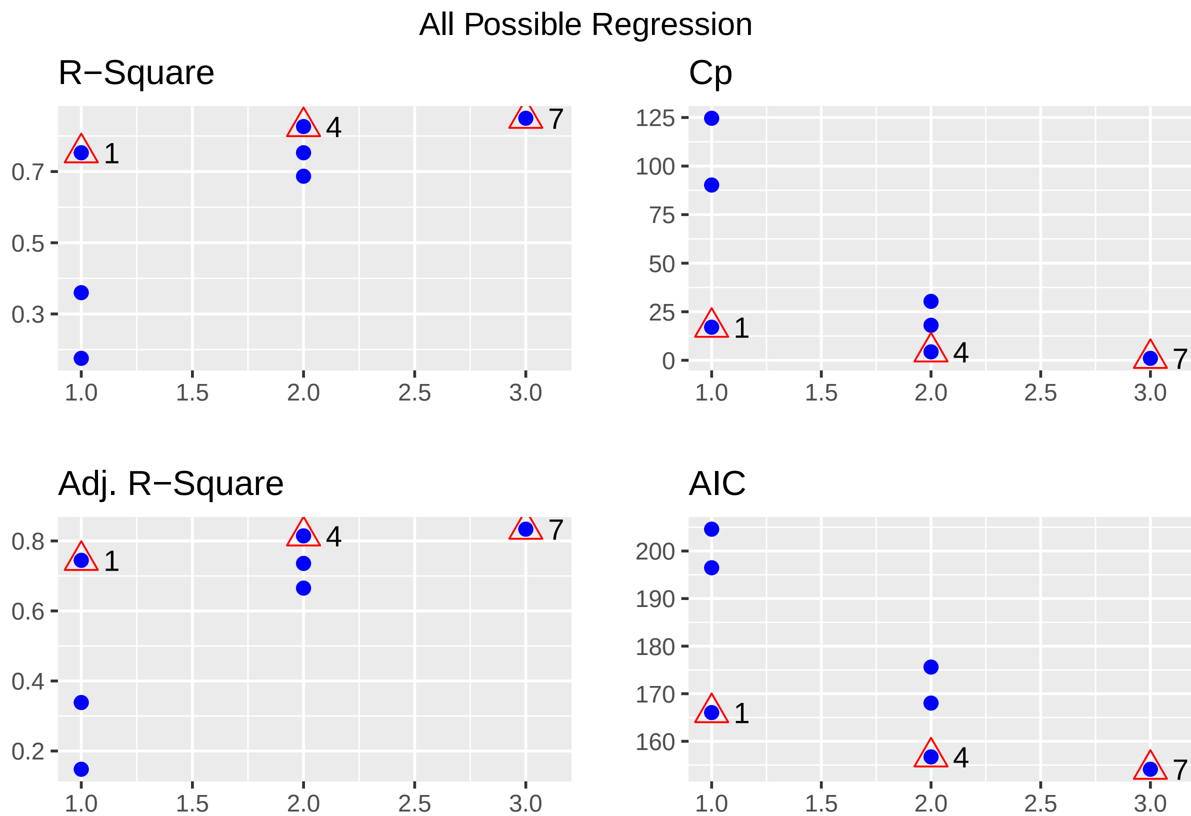
##Best Subset Regression

Select the subset of predictors that do the best at meeting some well-defined objective criterion, such as having the largest R2 value or the smallest MSE, Mallow's Cp or AIC.

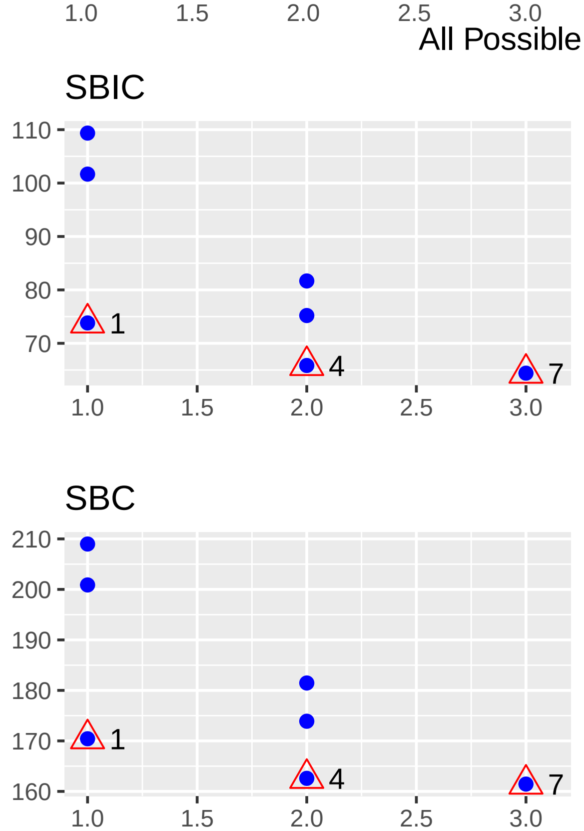
```
library(olsrr)
library(datasets)
exp <- lm(mpg ~wt+ qsec+ am, data = mtcars)
k <- ols_step_all_possible(exp)
k
```

```
##   Index N Predictors  R-Square Adj. R-Square Mallow's Cp
## 1     1 1         wt 0.7528328     0.7445939   18.034625
## 3     2 1         am 0.3597989     0.3384589   91.236754
## 2     3 1       qsec 0.1752963     0.1478062  125.600168
## 4     4 2    wt qsec 0.8264161     0.8144448    6.329809
## 5     5 2      wt am 0.7528348     0.7357889   20.034254
## 6     6 2    qsec am 0.6868398     0.6652425   32.325753
## 7     7 3 wt qsec am 0.8496636     0.8335561    4.000000
```

```r
plot(k)
```



All Possible Regression
R−Square
Cp
Adj. R−Square
AIC
1
4
7

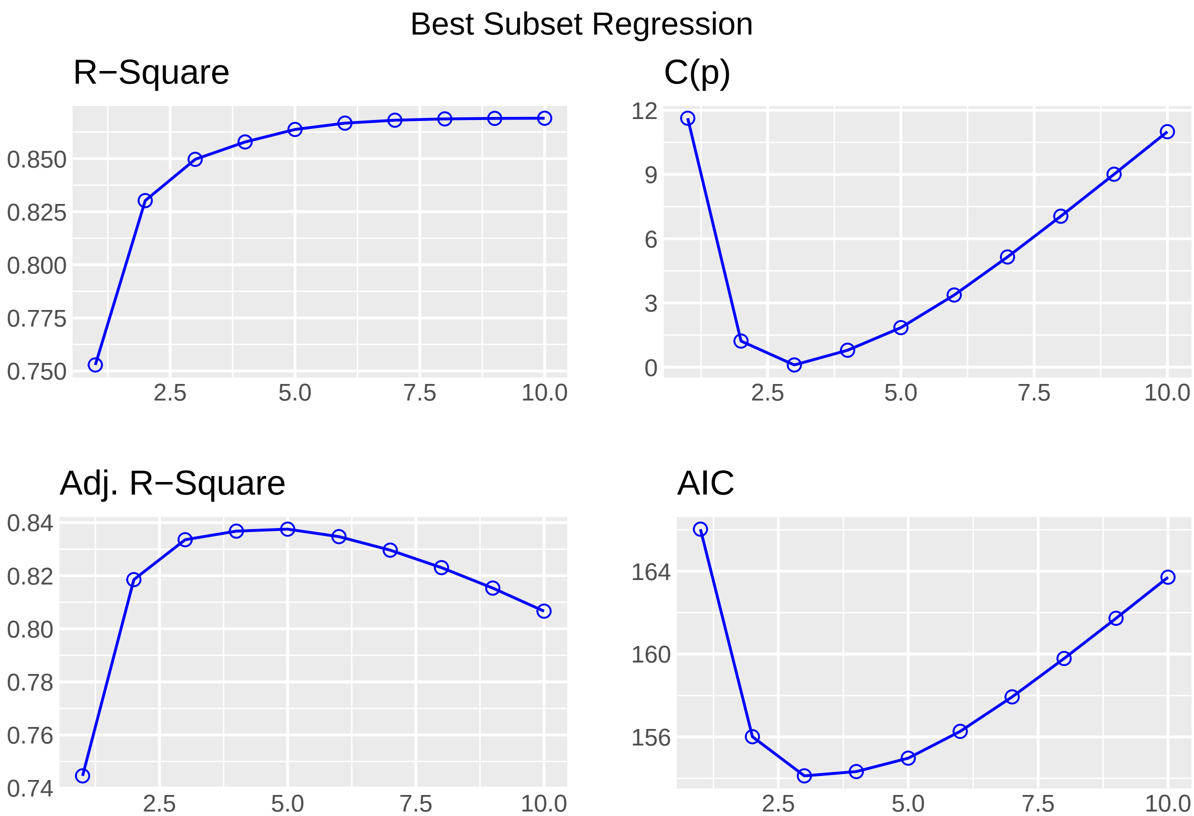


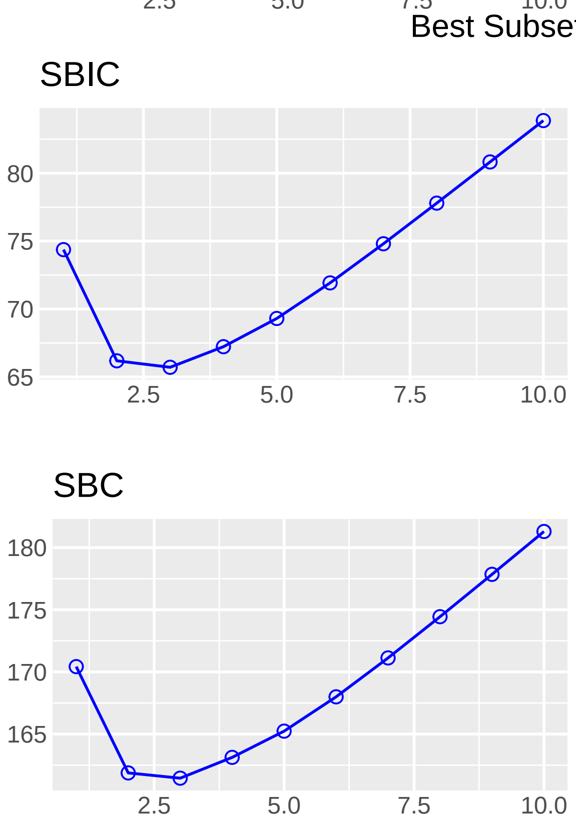
All Possible Regression
SBIC
SBC
1
4
7

```r
#lets get the best model for each subset
exp1 <- lm(mpg ~., data = mtcars)
k1<-ols_step_best_subset(exp1)
k1
```

```
##                   Best Subsets Regression
## ------------------------------------------------------
## Model Index    Predictors
## ------------------------------------------------------
##      1         wt
##      2         cyl wt
##      3         wt qsec am
##      4         hp wt qsec am
##      5         disp hp wt qsec am
##      6         disp hp drat wt qsec am
##      7         disp hp drat wt qsec am gear
##      8         disp hp drat wt qsec am gear carb
##      9         disp hp drat wt qsec vs am gear carb
##     10         cyl disp hp drat wt qsec vs am gear carb
## ------------------------------------------------------
##
##                                                    Subsets Regression Summary
## ----------------------------------------------------------------------------------------------------
##                        Adj.        Pred
## Model    R-Square    R-Square    R-Square     C(p)        AIC         SBIC        SBC         MSEP
## ----------------------------------------------------------------------------------------------------
##   1        0.7528      0.7446      0.7087    11.6270    166.0294     74.3734    170.4266    296.9167
##   2        0.8302      0.8185      0.7904     1.2187    156.0101     66.1903    161.8730    211.2280
##   3        0.8497      0.8336      0.7946     0.1026    154.1194     65.7138    161.4481    193.9735
##   4        0.8579      0.8368      0.8021     0.7900    154.3274     67.2299    163.1218    190.4637
##   5        0.8637      0.8375      0.7984     1.8462    154.9740     69.3073    165.2341    189.8793
##   6        0.8667      0.8347      0.7855     3.3700    156.2687     71.9258    167.9946    193.4796
##   7        0.8681      0.8296      0.7619     5.1472    157.9333     74.8058    171.1250    199.7867
##   8        0.8687      0.8230      0.7316     7.0496    159.7853     77.7959    174.4427    207.9040
##   9        0.8689      0.8153      0.7035     9.0114    161.7271     80.8277    177.8502    217.4086
##  10        0.8690      0.8066      0.6538    11.0000    163.7098     83.8728    181.2986    228.1554
## ----------------------------------------------------------------------------------------------------
## AIC: Akaike Information Criteria
##  SBIC: Sawa's Bayesian Information Criteria
##  SBC: Schwarz Bayesian Criteria
##  MSEP: Estimated error of prediction, assuming multivariate normality
##  FPE: Final Prediction Error
##  HSP: Hocking's Sp
##  APC: Amemiya Prediction Criteria
```

```r
plot(k1)
```



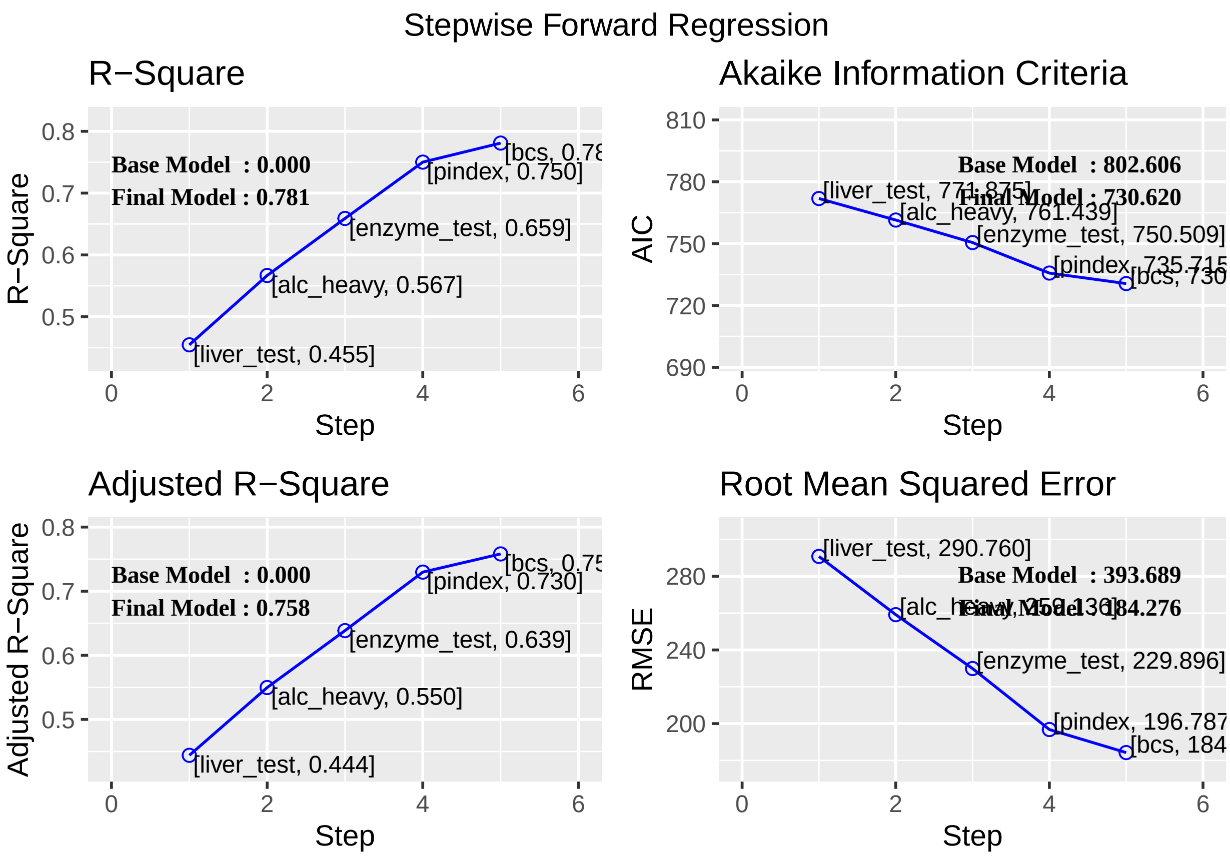


The plot method shows the panel of fit criteria for best subset regression methods.

##Stepwise Forward Regression

Build regression model from a set of candidate predictor variables by entering predictors based on p values, in a stepwise manner until there is no variable left to enter any more. The model should include all the candidate predictor variables. If details is set to TRUE, each step is displayed.

```
# stepwise forward regression
exp2 <- lm(y ~ ., data = surgical)
k2<-ols_step_forward_p(exp2)
plot(k2)
```




```
# stepwise forward regression
ols_step_forward_p(exp2, details = TRUE)
```

```
## Forward Selection Method
## ------------------------
##
## Candidate Terms:
##
## 1. bcs
## 2. pindex
## 3. enzyme_test
## 4. liver_test
## 5. age
## 6. gender
## 7. alc_mod
## 8. alc_heavy
##
##
## Step   => 0
```

```
## Model  => y ~ 1
## R2     => 0
##
## Initiating stepwise selection...
##
##                      Selection Metrics Table
## -----------------------------------------------------------------
## Predictor      Pr(>|t|)    R-Squared    Adj. R-Squared      AIC
## -----------------------------------------------------------------
## liver_test      0.00000        0.455             0.444    771.875
## enzyme_test     0.00000        0.334             0.322    782.629
## pindex          0.00155        0.177             0.161    794.100
## alc_heavy       0.00172        0.174             0.158    794.301
## bcs             0.01025        0.120             0.103    797.697
## alc_mod         0.19286        0.032             0.014    802.828
## gender          0.20972        0.030             0.011    802.956
## age             0.39073        0.014            -0.005    803.834
## -----------------------------------------------------------------
##
## Step      => 1
## Selected  => liver_test
## Model     => y ~ liver_test
## R2        => 0.455
##
##                      Selection Metrics Table
## -----------------------------------------------------------------
## Predictor      Pr(>|t|)    R-Squared    Adj. R-Squared      AIC
## -----------------------------------------------------------------
## alc_heavy       0.00065        0.567             0.550    761.439
## enzyme_test     0.00089        0.562             0.544    762.077
## pindex          0.07087        0.489             0.469    770.387
## alc_mod         0.10979        0.481             0.461    771.141
## gender          0.79395        0.455             0.434    773.802
## age             0.83908        0.455             0.434    773.831
## bcs             0.93062        0.455             0.433    773.867
## -----------------------------------------------------------------
##
## Step      => 2
## Selected  => alc_heavy
## Model     => y ~ liver_test + alc_heavy
## R2        => 0.567
##
##                      Selection Metrics Table
## -----------------------------------------------------------------
## Predictor      Pr(>|t|)    R-Squared    Adj. R-Squared      AIC
## -----------------------------------------------------------------
## enzyme_test     0.00057        0.659             0.639    750.509
## pindex          0.00961        0.622             0.599    756.125
## bcs             0.55687        0.570             0.544    763.063
## age             0.58269        0.569             0.544    763.110
## alc_mod         0.91757        0.567             0.541    763.428
## gender          0.93799        0.567             0.541    763.433
## -----------------------------------------------------------------
##
```

```
## Step      => 3
## Selected  => enzyme_test
## Model     => y ~ liver_test + alc_heavy + enzyme_test
## R2        => 0.659
##
##                     Selection Metrics Table
## ---------------------------------------------------------------
## Predictor    Pr(>|t|)    R-Squared    Adj. R-Squared      AIC
## ---------------------------------------------------------------
## pindex          1e-04        0.750             0.730    735.715
## bcs           0.21294        0.670             0.643    750.782
## alc_mod       0.75743        0.660             0.632    752.403
## age           0.77290        0.660             0.632    752.416
## gender        0.99197        0.659             0.631    752.509
## ---------------------------------------------------------------
##
## Step      => 4
## Selected  => pindex
## Model     => y ~ liver_test + alc_heavy + enzyme_test + pindex
## R2        => 0.75
##
##                     Selection Metrics Table
## ---------------------------------------------------------------
## Predictor    Pr(>|t|)    R-Squared    Adj. R-Squared      AIC
## ---------------------------------------------------------------
## bcs           0.01248        0.781             0.758    730.620
## age           0.86220        0.750             0.724    737.680
## gender        0.96390        0.750             0.724    737.712
## alc_mod       0.97040        0.750             0.724    737.713
## ---------------------------------------------------------------
##
## Step      => 5
## Selected  => bcs
## Model     => y ~ liver_test + alc_heavy + enzyme_test + pindex + bcs
## R2        => 0.781
##
##                     Selection Metrics Table
## ---------------------------------------------------------------
## Predictor    Pr(>|t|)    R-Squared    Adj. R-Squared      AIC
## ---------------------------------------------------------------
## age           0.74164        0.781             0.754    732.494
## gender        0.80666        0.781             0.753    732.551
## alc_mod       0.94086        0.781             0.753    732.614
## ---------------------------------------------------------------
##
##
## No more variables to be added.
##
## Variables Selected:
##
## => liver_test
## => alc_heavy
## => enzyme_test
## => pindex
```

```
## => bcs
##
##
##                               Stepwise Summary
## -------------------------------------------------------------------------
## Step    Variable        AIC        SBC       SBIC        R2       Adj. R2
## -------------------------------------------------------------------------
##  0      Base Model    802.606    806.584    646.794    0.00000    0.00000
##  1      liver_test    771.875    777.842    616.009    0.45454    0.44405
##  2      alc_heavy     761.439    769.395    605.506    0.56674    0.54975
##  3      enzyme_test   750.509    760.454    595.297    0.65900    0.63854
##  4      pindex        735.715    747.649    582.943    0.75015    0.72975
##  5      bcs           730.620    744.543    579.638    0.78091    0.75808
## -------------------------------------------------------------------------
##
## Final Model Output
## ------------------
##
##                            Model Summary
## ------------------------------------------------------------------
## R                         0.884       RMSE                 184.276
## R-Squared                 0.781       MSE                33957.712
## Adj. R-Squared            0.758       Coef. Var             27.839
## Pred R-Squared            0.700       AIC                  730.620
## MAE                     137.656       SBC                  744.543
## ------------------------------------------------------------------
##  RMSE: Root Mean Square Error
##  MSE: Mean Square Error
##  MAE: Mean Absolute Error
##  AIC: Akaike Information Criteria
##  SBC: Schwarz Bayesian Criteria
##
##                                   ANOVA
## ----------------------------------------------------------------------
##                   Sum of
##                  Squares        DF    Mean Square      F         Sig.
## ----------------------------------------------------------------------
## Regression    6535804.090        5    1307160.818    34.217    0.0000
## Residual      1833716.447       48      38202.426
## Total         8369520.537       53
## ----------------------------------------------------------------------
##
##                                        Parameter Estimates
## -----------------------------------------------------------------------------------------------
##       model         Beta    Std. Error    Std. Beta      t        Sig         lower       upper
## -----------------------------------------------------------------------------------------------
## (Intercept)    -1178.330       208.682                 -5.647    0.000    -1597.914    -758.746
##  liver_test       58.064        40.144        0.156     1.446    0.155      -22.652     138.779
##   alc_heavy      317.848        71.634        0.314     4.437    0.000      173.818     461.878
## enzyme_test        9.748         1.656        0.521     5.887    0.000        6.419      13.077
##      pindex        8.924         1.808        0.380     4.935    0.000        5.288      12.559
##         bcs       59.864        23.060        0.241     2.596    0.012       13.498     106.230
## -----------------------------------------------------------------------------------------------
```

##Stepwise Backward Regression

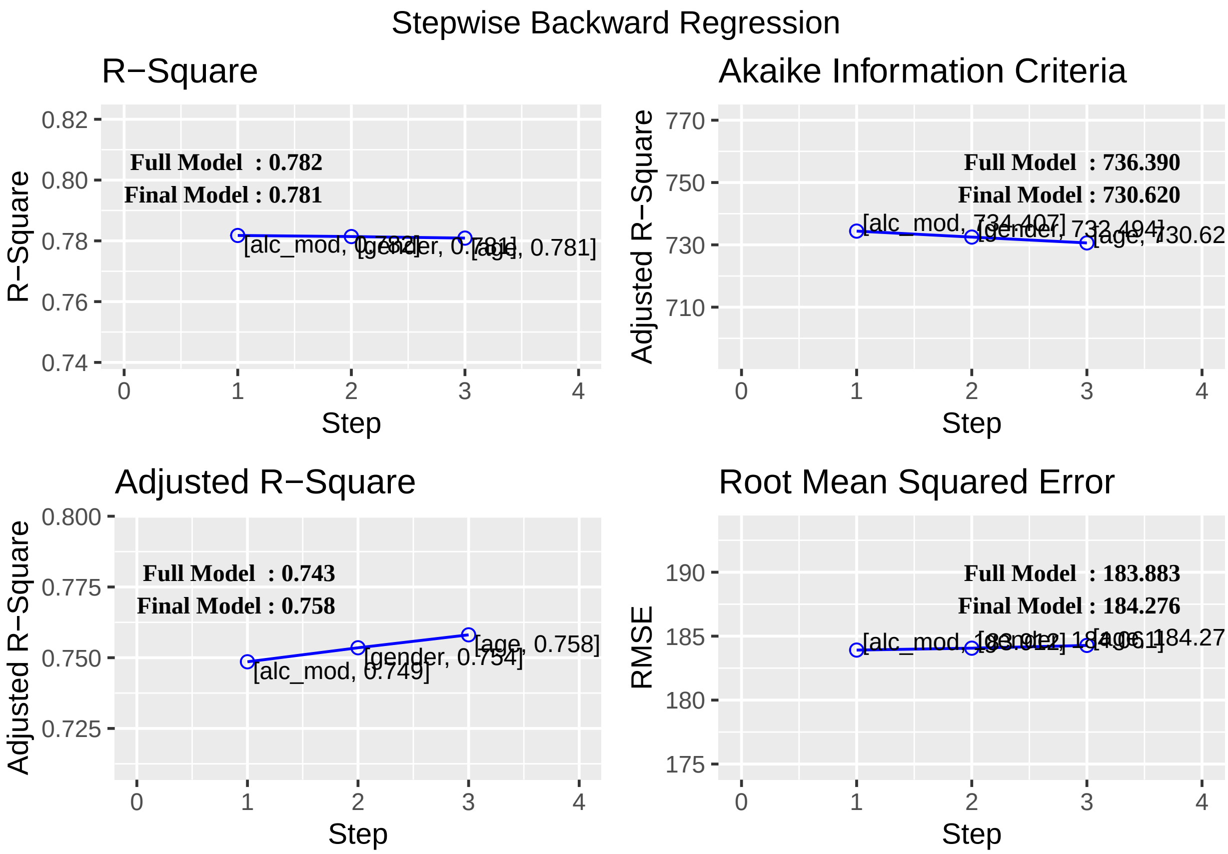
Build regression model from a set of candidate predictor variables by removing predictors based on p values, in a stepwise manner until there is no variable left to remove any more. The model should include all the candidate predictor variables. If details is set to TRUE, each step is displayed.

```
# stepwise backward regression
k4<-ols_step_backward_p(exp2)
k4
```

```
## 
## 
##                               Stepwise Summary                              
## --------------------------------------------------------------------------
## Step    Variable        AIC        SBC       SBIC        R2       Adj. R2  
## --------------------------------------------------------------------------
##  0      Full Model    736.390    756.280    586.665    0.78184    0.74305  
##  1      alc_mod       734.407    752.308    584.276    0.78177    0.74856  
##  2      gender        732.494    748.406    581.938    0.78142    0.75351  
##  3      age           730.620    744.543    579.638    0.78091    0.75808  
## --------------------------------------------------------------------------
## 
## Final Model Output 
## ------------------
## 
##                            Model Summary                              
## -------------------------------------------------------------------
## R                          0.884       RMSE                184.276 
## R-Squared                  0.781       MSE               33957.712 
## Adj. R-Squared             0.758       Coef. Var            27.839 
## Pred R-Squared             0.700       AIC                 730.620 
## MAE                      137.656       SBC                 744.543 
## -------------------------------------------------------------------
##  RMSE: Root Mean Square Error 
##  MSE: Mean Square Error 
##  MAE: Mean Absolute Error 
##  AIC: Akaike Information Criteria 
##  SBC: Schwarz Bayesian Criteria 
## 
##                                  ANOVA                                   
## -----------------------------------------------------------------------
##                     Sum of                                             
##                    Squares        DF    Mean Square      F         Sig. 
## -----------------------------------------------------------------------
## Regression    6535804.090         5    1307160.818    34.217    0.0000 
## Residual      1833716.447        48      38202.426                      
## Total         8369520.537        53                                     
## -----------------------------------------------------------------------
## 
##                                        Parameter Estimates                                        
## -------------------------------------------------------------------------------------------------
##       model          Beta    Std. Error    Std. Beta      t        Sig         lower        upper 
## -------------------------------------------------------------------------------------------------
## (Intercept)    -1178.330       208.682                 -5.647    0.000    -1597.914     -758.746 
##         bcs       59.864        23.060        0.241     2.596    0.012       13.498      106.230 
```

```
##      pindex        8.924          1.808          0.380      4.935     0.000          5.288        12.559
## enzyme_test        9.748          1.656          0.521      5.887     0.000          6.419        13.077
##  liver_test       58.064         40.144          0.156      1.446     0.155        -22.652       138.779
##   alc_heavy      317.848         71.634          0.314      4.437     0.000        173.818       461.878
## ----------------------------------------------------------------------------------------------------
```

```r
plot(k4)
```




```r
#get the model
k4$model
```

```
##
## Call:
## lm(formula = paste(response, "~", paste(c(include, cterms), collapse = " + ")),
##     data = l)
##
## Coefficients:
## (Intercept)          bcs       pindex  enzyme_test   liver_test    alc_heavy
##   -1178.330       59.864        8.924        9.748       58.064      317.848
```

```r
formula(k4$model)
```

```
## y ~ bcs + pindex + enzyme_test + liver_test + alc_heavy
## <environment: 0x591d6ab9d948>
```

```r
#Detailed Output
# stepwise backward regression
ols_step_backward_p(exp2, details = TRUE)
```

```
## Backward Elimination Method
```

```
## --------------------------
##
## Candidate Terms:
##
## 1. bcs
## 2. pindex
## 3. enzyme_test
## 4. liver_test
## 5. age
## 6. gender
## 7. alc_mod
## 8. alc_heavy
##
##
## Step   => 0
## Model  => y ~ bcs + pindex + enzyme_test + liver_test + age + gender + alc_mod + alc_heavy
## R2     => 0.782
##
## Initiating stepwise selection...
##
## Step     => 1
## Removed  => alc_mod
## Model    => y ~ bcs + pindex + enzyme_test + liver_test + age + gender + alc_heavy
## R2       => 0.78177
##
## Step     => 2
## Removed  => gender
## Model    => y ~ bcs + pindex + enzyme_test + liver_test + age + alc_heavy
## R2       => 0.78142
##
## Step     => 3
## Removed  => age
## Model    => y ~ bcs + pindex + enzyme_test + liver_test + alc_heavy
## R2       => 0.78091
##
##
## No more variables to be removed.
##
## Variables Removed:
##
## => alc_mod
## => gender
## => age

##
##
##                             Stepwise Summary
## ---------------------------------------------------------------------------
## Step    Variable        AIC        SBC       SBIC        R2       Adj. R2
## ---------------------------------------------------------------------------
##  0      Full Model    736.390    756.280    586.665    0.78184    0.74305
##  1      alc_mod       734.407    752.308    584.276    0.78177    0.74856
##  2      gender        732.494    748.406    581.938    0.78142    0.75351
##  3      age           730.620    744.543    579.638    0.78091    0.75808
```

```
## -------------------------------------------------------------------------
##
## Final Model Output
## ------------------
##
##                            Model Summary
## -------------------------------------------------------------------
## R                          0.884       RMSE                184.276
## R-Squared                  0.781       MSE               33957.712
## Adj. R-Squared             0.758       Coef. Var            27.839
## Pred R-Squared             0.700       AIC                 730.620
## MAE                      137.656       SBC                 744.543
## -------------------------------------------------------------------
##  RMSE: Root Mean Square Error
##  MSE: Mean Square Error
##  MAE: Mean Absolute Error
##  AIC: Akaike Information Criteria
##  SBC: Schwarz Bayesian Criteria
##
##                                  ANOVA
## -----------------------------------------------------------------------
##                   Sum of
##                  Squares        DF    Mean Square      F         Sig.
## -----------------------------------------------------------------------
## Regression    6535804.090         5    1307160.818    34.217    0.0000
## Residual      1833716.447        48      38202.426
## Total         8369520.537        53
## -----------------------------------------------------------------------
##
##                                       Parameter Estimates
## -------------------------------------------------------------------------------------------------
##       model          Beta    Std. Error    Std. Beta      t         Sig          lower        upper
## -------------------------------------------------------------------------------------------------
## (Intercept)    -1178.330       208.682                 -5.647    0.000    -1597.914     -758.746
##         bcs       59.864        23.060        0.241     2.596    0.012       13.498      106.230
##      pindex        8.924         1.808        0.380     4.935    0.000        5.288       12.559
## enzyme_test        9.748         1.656        0.521     5.887    0.000        6.419       13.077
##  liver_test       58.064        40.144        0.156     1.446    0.155      -22.652      138.779
##   alc_heavy      317.848        71.634        0.314     4.437    0.000      173.818      461.878
## -------------------------------------------------------------------------------------------------
```

##Stepwise Regression Build regression model from a set of candidate predictor variables by entering and removing predictors based on p values, in a stepwise manner until there is no variable left to enter or remove any more. The model should include all the candidate predictor variables. If details is set to TRUE, each step is displayed.

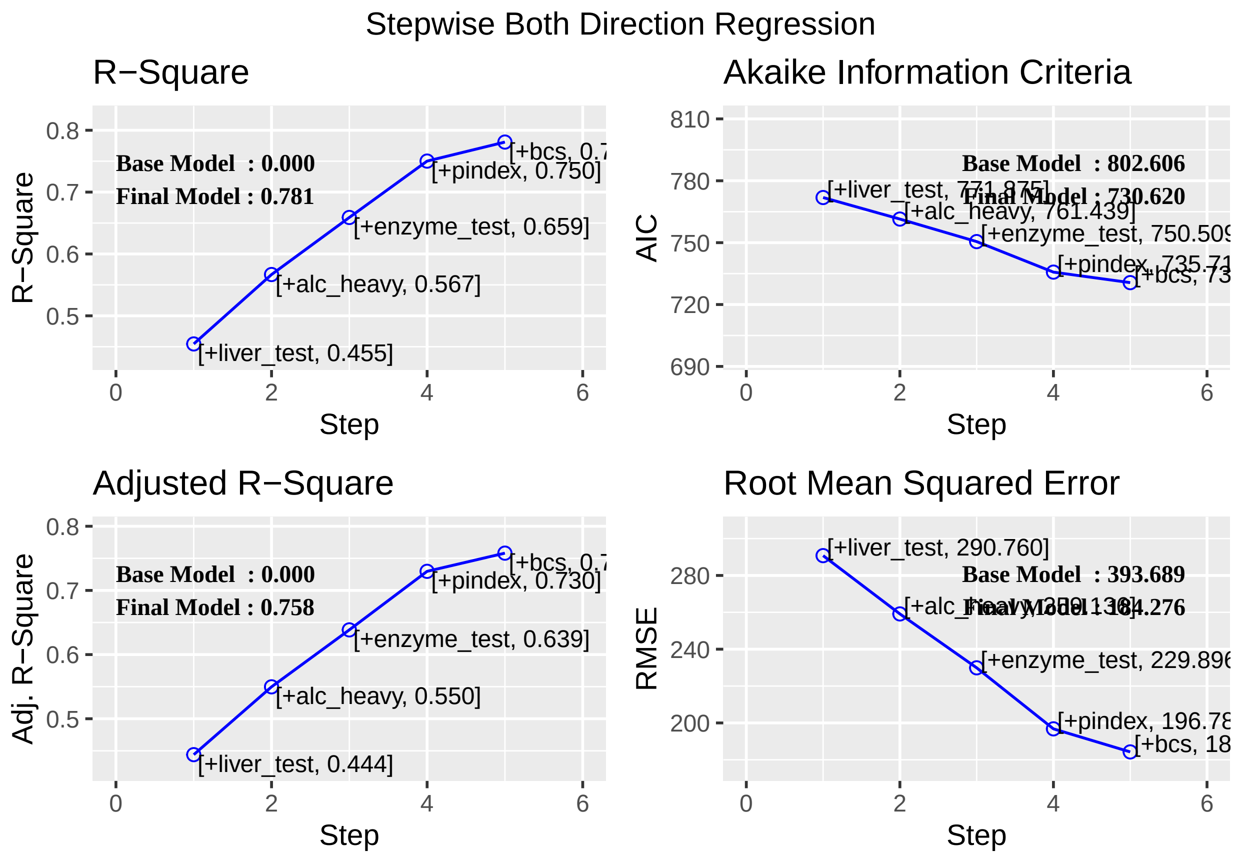
```r
# stepwise regression

k5<-ols_step_both_p(exp2)
k5
```

```
##
##
##                                 Stepwise Summary
## ------------------------------------------------------------------------------
```

```
## Step    Variable            AIC       SBC       SBIC        R2       Adj. R2
## --------------------------------------------------------------------------------
##  0      Base Model         802.606   806.584   646.794    0.00000    0.00000
##  1      liver_test (+)     771.875   777.842   616.009    0.45454    0.44405
##  2      alc_heavy (+)      761.439   769.395   605.506    0.56674    0.54975
##  3      enzyme_test (+)    750.509   760.454   595.297    0.65900    0.63854
##  4      pindex (+)         735.715   747.649   582.943    0.75015    0.72975
##  5      bcs (+)            730.620   744.543   579.638    0.78091    0.75808
## --------------------------------------------------------------------------------
##
## Final Model Output
## ------------------
##
##                           Model Summary
## ---------------------------------------------------------------------
## R                         0.884       RMSE                  184.276
## R-Squared                 0.781       MSE                 33957.712
## Adj. R-Squared            0.758       Coef. Var              27.839
## Pred R-Squared            0.700       AIC                   730.620
## MAE                     137.656       SBC                   744.543
## ---------------------------------------------------------------------
##  RMSE: Root Mean Square Error
##  MSE: Mean Square Error
##  MAE: Mean Absolute Error
##  AIC: Akaike Information Criteria
##  SBC: Schwarz Bayesian Criteria
##
##                                 ANOVA
## -------------------------------------------------------------------------
##                  Sum of
##                 Squares        DF    Mean Square      F         Sig.
## -------------------------------------------------------------------------
## Regression    6535804.090         5    1307160.818    34.217    0.0000
## Residual      1833716.447        48      38202.426
## Total         8369520.537        53
## -------------------------------------------------------------------------
##
##                                      Parameter Estimates
## ----------------------------------------------------------------------------------------------
##       model         Beta    Std. Error    Std. Beta      t        Sig        lower       upper
## ----------------------------------------------------------------------------------------------
## (Intercept)    -1178.330       208.682                 -5.647    0.000    -1597.914    -758.746
##  liver_test       58.064        40.144        0.156     1.446    0.155      -22.652     138.779
##   alc_heavy      317.848        71.634        0.314     4.437    0.000      173.818     461.878
## enzyme_test        9.748         1.656        0.521     5.887    0.000        6.419      13.077
##      pindex        8.924         1.808        0.380     4.935    0.000        5.288      12.559
##         bcs       59.864        23.060        0.241     2.596    0.012       13.498     106.230
## ----------------------------------------------------------------------------------------------
```

```r
plot(k5)
```




```r
#ensuring that all variables are significant at 95% level
k5.1<-ols_step_both_p(exp2,p_enter=0.1,p_remove=0.05,details=TRUE)
```

```
## Stepwise Selection Method
## -------------------------
##
## Candidate Terms:
##
## 1. bcs
## 2. pindex
## 3. enzyme_test
## 4. liver_test
## 5. age
## 6. gender
## 7. alc_mod
## 8. alc_heavy
##
##
## Step   => 0
## Model  => y ~ 1
## R2     => 0
##
## Initiating stepwise selection...
##
## Step      => 1
## Selected  => liver_test
## Model     => y ~ liver_test
```

```
## R2        => 0.455
##
## Step      => 2
## Selected  => alc_heavy
## Model     => y ~ liver_test + alc_heavy
## R2        => 0.567
##
## Step      => 3
## Selected  => enzyme_test
## Model     => y ~ liver_test + alc_heavy + enzyme_test
## R2        => 0.659
##
## Step      => 4
## Selected  => pindex
## Model     => y ~ liver_test + alc_heavy + enzyme_test + pindex
## R2        => 0.75
##
## Step      => 5
## Selected  => bcs
## Model     => y ~ liver_test + alc_heavy + enzyme_test + pindex + bcs
## R2        => 0.781
##
## Step     => 6
## Removed  => liver_test
## Model    => y ~ alc_heavy + enzyme_test + pindex + bcs
##   => 0.77136
##
##
## No more variables to be added or removed.
```

```
k5.1
```

```
##
##
##                                 Stepwise Summary
## -------------------------------------------------------------------------------
## Step    Variable            AIC        SBC       SBIC        R2       Adj. R2
## -------------------------------------------------------------------------------
##  0      Base Model        802.606    806.584    646.794    0.00000    0.00000
##  1      liver_test (+)    771.875    777.842    616.009    0.45454    0.44405
##  2      alc_heavy (+)     761.439    769.395    605.506    0.56674    0.54975
##  3      enzyme_test (+)   750.509    760.454    595.297    0.65900    0.63854
##  4      pindex (+)        735.715    747.649    582.943    0.75015    0.72975
##  5      bcs (+)           730.620    744.543    579.638    0.78091    0.75808
##  6      liver_test (-)    730.924    742.858    579.087    0.77136    0.75269
## -------------------------------------------------------------------------------
##
## Final Model Output
## ------------------
##
##                            Model Summary
## ------------------------------------------------------------------
## R                         0.878       RMSE                 188.249
## R-Squared                 0.771       MSE                35437.709
## Adj. R-Squared            0.753       Coef. Var             28.147
```

```
## Pred R-Squared             0.695       AIC                  730.924
## MAE                      140.619       SBC                  742.858
## -------------------------------------------------------------------
##  RMSE: Root Mean Square Error
##  MSE: Mean Square Error
##  MAE: Mean Absolute Error
##  AIC: Akaike Information Criteria
##  SBC: Schwarz Bayesian Criteria
##
##                                  ANOVA
## -----------------------------------------------------------------------
##                    Sum of
##                   Squares        DF    Mean Square      F         Sig.
## -----------------------------------------------------------------------
## Regression    6455884.265         4    1613971.066    41.327    0.0000
## Residual      1913636.272        49      39053.801
## Total         8369520.537        53
## -----------------------------------------------------------------------
##
##                                       Parameter Estimates
## -----------------------------------------------------------------------------------------------
##       model         Beta    Std. Error    Std. Beta      t        Sig        lower        upper
## -----------------------------------------------------------------------------------------------
## (Intercept)    -1334.424       180.589                 -7.389    0.000    -1697.332     -971.516
##   alc_heavy      312.777        72.341        0.309     4.324    0.000      167.402      458.152
## enzyme_test       11.243         1.308        0.601     8.596    0.000        8.614       13.871
##      pindex       10.131         1.622        0.431     6.246    0.000        6.871       13.390
##         bcs       81.439        17.781        0.329     4.580    0.000       45.706      117.171
## -----------------------------------------------------------------------------------------------
```

```r
#Detailed Output
ols_step_both_p(exp2, details = TRUE)
```

```
## Stepwise Selection Method
## -------------------------
##
## Candidate Terms:
##
## 1. bcs
## 2. pindex
## 3. enzyme_test
## 4. liver_test
## 5. age
## 6. gender
## 7. alc_mod
## 8. alc_heavy
##
##
## Step   => 0
## Model  => y ~ 1
## R2     => 0
##
## Initiating stepwise selection...
##
## Step      => 1
```

```
## Selected  => liver_test
## Model     => y ~ liver_test
## R2        => 0.455
##
## Step      => 2
## Selected  => alc_heavy
## Model     => y ~ liver_test + alc_heavy
## R2        => 0.567
##
## Step      => 3
## Selected  => enzyme_test
## Model     => y ~ liver_test + alc_heavy + enzyme_test
## R2        => 0.659
##
## Step      => 4
## Selected  => pindex
## Model     => y ~ liver_test + alc_heavy + enzyme_test + pindex
## R2        => 0.75
##
## Step      => 5
## Selected  => bcs
## Model     => y ~ liver_test + alc_heavy + enzyme_test + pindex + bcs
## R2        => 0.781
##
##
## No more variables to be added or removed.
```

```
##
##
##                                  Stepwise Summary
## ------------------------------------------------------------------------------
## Step    Variable            AIC        SBC       SBIC        R2       Adj. R2
## ------------------------------------------------------------------------------
##  0      Base Model        802.606    806.584    646.794    0.00000    0.00000
##  1      liver_test (+)    771.875    777.842    616.009    0.45454    0.44405
##  2      alc_heavy (+)     761.439    769.395    605.506    0.56674    0.54975
##  3      enzyme_test (+)   750.509    760.454    595.297    0.65900    0.63854
##  4      pindex (+)        735.715    747.649    582.943    0.75015    0.72975
##  5      bcs (+)           730.620    744.543    579.638    0.78091    0.75808
## ------------------------------------------------------------------------------
##
## Final Model Output
## ------------------
##
##                             Model Summary
## -------------------------------------------------------------------
## R                         0.884       RMSE                 184.276
## R-Squared                 0.781       MSE                33957.712
## Adj. R-Squared            0.758       Coef. Var             27.839
## Pred R-Squared            0.700       AIC                  730.620
## MAE                     137.656       SBC                  744.543
## -------------------------------------------------------------------
##  RMSE: Root Mean Square Error
##  MSE: Mean Square Error
```

```
##  MAE: Mean Absolute Error
##  AIC: Akaike Information Criteria
##  SBC: Schwarz Bayesian Criteria
##
##                                  ANOVA
## ----------------------------------------------------------------------
##                     Sum of
##                    Squares        DF    Mean Square      F         Sig.
## ----------------------------------------------------------------------
## Regression    6535804.090         5    1307160.818    34.217    0.0000
## Residual      1833716.447        48      38202.426
## Total         8369520.537        53
## ----------------------------------------------------------------------
##
##                                       Parameter Estimates
## ----------------------------------------------------------------------------------------------------
##       model         Beta    Std. Error    Std. Beta       t        Sig        lower        upper
## ----------------------------------------------------------------------------------------------------
## (Intercept)    -1178.330       208.682                 -5.647    0.000    -1597.914     -758.746
##  liver_test       58.064        40.144        0.156     1.446    0.155      -22.652      138.779
##   alc_heavy      317.848        71.634        0.314     4.437    0.000      173.818      461.878
## enzyme_test        9.748         1.656        0.521     5.887    0.000        6.419       13.077
##      pindex        8.924         1.808        0.380     4.935    0.000        5.288       12.559
##         bcs       59.864        23.060        0.241     2.596    0.012       13.498      106.230
## ----------------------------------------------------------------------------------------------------
```

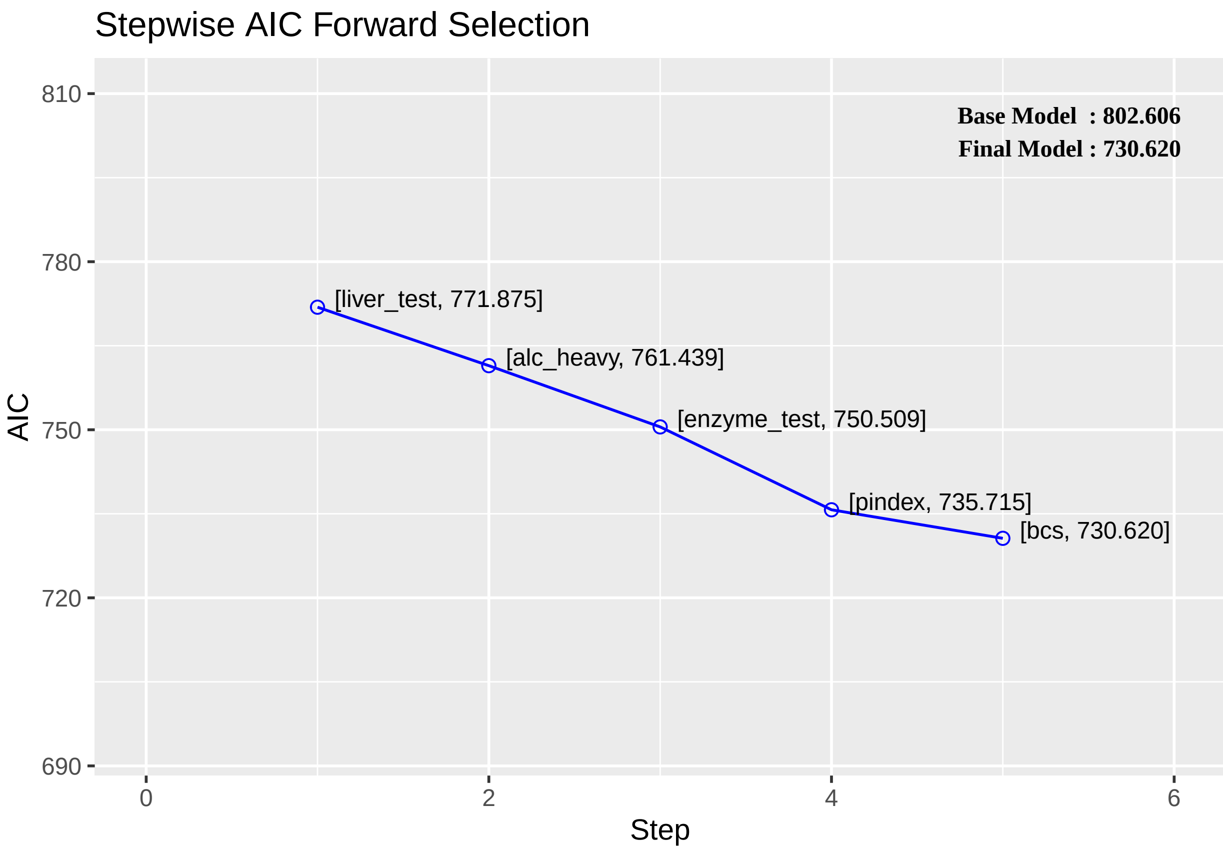
##Stepwise AIC Forward Regression Build regression model from a set of candidate predictor variables by entering predictors based on Akaike Information Criteria, in a stepwise manner until there is no variable left to enter any more. The model should include all the candidate predictor variables. If details is set to TRUE, each step is displayed.

```r
# stepwise aic forward regression
k6<-ols_step_forward_aic(exp2)
k6
```

```
##
##
##                                Stepwise Summary
## ---------------------------------------------------------------------------
## Step    Variable         AIC        SBC        SBIC        R2        Adj. R2
## ---------------------------------------------------------------------------
##  0      Base Model     802.606    806.584    646.794    0.00000    0.00000
##  1      liver_test     771.875    777.842    616.009    0.45454    0.44405
##  2      alc_heavy      761.439    769.395    605.506    0.56674    0.54975
##  3      enzyme_test    750.509    760.454    595.297    0.65900    0.63854
##  4      pindex         735.715    747.649    582.943    0.75015    0.72975
##  5      bcs            730.620    744.543    579.638    0.78091    0.75808
## ---------------------------------------------------------------------------
##
## Final Model Output
## ------------------
##
##                             Model Summary
## ------------------------------------------------------------------
## R                           0.884       RMSE                184.276
```

```
## R-Squared                  0.781       MSE                33957.712
## Adj. R-Squared             0.758       Coef. Var             27.839
## Pred R-Squared             0.700       AIC                  730.620
## MAE                      137.656       SBC                  744.543
## ------------------------------------------------------------------
##  RMSE: Root Mean Square Error
##  MSE: Mean Square Error
##  MAE: Mean Absolute Error
##  AIC: Akaike Information Criteria
##  SBC: Schwarz Bayesian Criteria
##
##                                ANOVA
## ----------------------------------------------------------------------
##                    Sum of
##                   Squares        DF    Mean Square      F         Sig.
## ----------------------------------------------------------------------
## Regression    6535804.090         5    1307160.818    34.217    0.0000
## Residual      1833716.447        48      38202.426
## Total         8369520.537        53
## ----------------------------------------------------------------------
##
##                                     Parameter Estimates
## ------------------------------------------------------------------------------------------------
##       model         Beta    Std. Error    Std. Beta      t        Sig        lower       upper
## ------------------------------------------------------------------------------------------------
## (Intercept)    -1178.330       208.682                 -5.647    0.000    -1597.914    -758.746
##  liver_test       58.064        40.144        0.156     1.446    0.155      -22.652     138.779
##   alc_heavy      317.848        71.634        0.314     4.437    0.000      173.818     461.878
## enzyme_test        9.748         1.656        0.521     5.887    0.000        6.419      13.077
##      pindex        8.924         1.808        0.380     4.935    0.000        5.288      12.559
##         bcs       59.864        23.060        0.241     2.596    0.012       13.498     106.230
## ------------------------------------------------------------------------------------------------
```

```r
plot(k6)
```




```r
ols_step_forward_aic(exp2, details = TRUE)
```

```
## Forward Selection Method
## ------------------------
##
## Candidate Terms:
##
## 1. bcs
## 2. pindex
## 3. enzyme_test
## 4. liver_test
## 5. age
## 6. gender
## 7. alc_mod
## 8. alc_heavy
##
##
## Step     => 0
## Model    => y ~ 1
## AIC      => 802.606
##
## Initiating stepwise selection...
##
##                          Table: Adding New Variables
## -------------------------------------------------------------------------
## Predictor      DF      AIC        SBC        SBIC         R2        Adj. R2
## -------------------------------------------------------------------------
```

```
## liver_test      1    771.875    777.842    616.009    0.45454     0.44405
## enzyme_test     1    782.629    788.596    626.220    0.33435     0.32154
## pindex          1    794.100    800.067    637.196    0.17680     0.16097
## alc_heavy       1    794.301    800.268    637.389    0.17373     0.15784
## bcs             1    797.697    803.664    640.655    0.12010     0.10318
## alc_mod         1    802.828    808.795    645.601    0.03239     0.01378
## gender          1    802.956    808.923    645.725    0.03009     0.01143
## age             1    803.834    809.801    646.572    0.01420    -0.00476
## ----------------------------------------------------------------------------
##
## Step     => 1
## Added    => liver_test
## Model    => y ~ liver_test
## AIC      => 771.8753
##
##                        Table: Adding New Variables
## ---------------------------------------------------------------------------
## Predictor      DF      AIC        SBC       SBIC        R2       Adj. R2
## ---------------------------------------------------------------------------
## alc_heavy       1    761.439    769.395    605.506    0.56674    0.54975
## enzyme_test     1    762.077    770.033    606.090    0.56159    0.54440
## pindex          1    770.387    778.343    613.737    0.48866    0.46861
## alc_mod         1    771.141    779.097    614.435    0.48147    0.46113
## gender          1    773.802    781.758    616.901    0.45528    0.43391
## age             1    773.831    781.787    616.928    0.45498    0.43361
## bcs             1    773.867    781.823    616.961    0.45462    0.43323
## ---------------------------------------------------------------------------
##
## Step     => 2
## Added    => alc_heavy
## Model    => y ~ liver_test + alc_heavy
## AIC      => 761.4394
##
##                        Table: Adding New Variables
## ---------------------------------------------------------------------------
## Predictor      DF      AIC        SBC       SBIC        R2       Adj. R2
## ---------------------------------------------------------------------------
## enzyme_test     1    750.509    760.454    595.297    0.65900    0.63854
## pindex          1    756.125    766.070    600.225    0.62163    0.59892
## bcs             1    763.063    773.008    606.379    0.56975    0.54394
## age             1    763.110    773.055    606.421    0.56938    0.54354
## alc_mod         1    763.428    773.373    606.704    0.56683    0.54084
## gender          1    763.433    773.378    606.709    0.56679    0.54080
## ---------------------------------------------------------------------------
##
## Step     => 3
## Added    => enzyme_test
## Model    => y ~ liver_test + alc_heavy + enzyme_test
## AIC      => 750.5089
##
##                       Table: Adding New Variables
## -------------------------------------------------------------------------
## Predictor    DF      AIC        SBC       SBIC        R2       Adj. R2
## -------------------------------------------------------------------------
```

```
## pindex        1    735.715    747.649    582.943    0.75015    0.72975
## bcs           1    750.782    762.716    595.377    0.66973    0.64277
## alc_mod       1    752.403    764.337    596.743    0.65967    0.63189
## age           1    752.416    764.350    596.755    0.65959    0.63180
## gender        1    752.509    764.443    596.833    0.65900    0.63116
## ----------------------------------------------------------------------
##
## Step     => 4
## Added    => pindex
## Model    => y ~ liver_test + alc_heavy + enzyme_test + pindex
## AIC      => 735.7146
##
##                      Table: Adding New Variables
## ----------------------------------------------------------------------
## Predictor    DF      AIC        SBC       SBIC        R2       Adj. R2
## ----------------------------------------------------------------------
## bcs           1    730.620    744.543    579.638    0.78091    0.75808
## age           1    737.680    751.603    585.012    0.75030    0.72429
## gender        1    737.712    751.635    585.036    0.75016    0.72413
## alc_mod       1    737.713    751.636    585.037    0.75015    0.72413
## ----------------------------------------------------------------------
##
## Step     => 5
## Added    => bcs
## Model    => y ~ liver_test + alc_heavy + enzyme_test + pindex + bcs
## AIC      => 730.6204
##
##                      Table: Adding New Variables
## ----------------------------------------------------------------------
## Predictor    DF      AIC        SBC       SBIC        R2       Adj. R2
## ----------------------------------------------------------------------
## age           1    732.494    748.406    581.938    0.78142    0.75351
## gender        1    732.551    748.463    581.978    0.78119    0.75325
## alc_mod       1    732.614    748.526    582.023    0.78093    0.75297
## ----------------------------------------------------------------------
##
##
## No more variables to be added.
##
## Variables Selected:
##
## => liver_test
## => alc_heavy
## => enzyme_test
## => pindex
## => bcs

##
##
##                               Stepwise Summary
## --------------------------------------------------------------------------
## Step    Variable         AIC        SBC       SBIC        R2       Adj. R2
## --------------------------------------------------------------------------
##  0      Base Model     802.606    806.584    646.794    0.00000    0.00000
```

```
## 1      liver_test     771.875    777.842    616.009    0.45454    0.44405
## 2      alc_heavy      761.439    769.395    605.506    0.56674    0.54975
## 3      enzyme_test    750.509    760.454    595.297    0.65900    0.63854
## 4      pindex         735.715    747.649    582.943    0.75015    0.72975
## 5      bcs            730.620    744.543    579.638    0.78091    0.75808
## -------------------------------------------------------------------------
##
## Final Model Output
## ------------------
##
##                            Model Summary
## ------------------------------------------------------------------
## R                        0.884       RMSE               184.276
## R-Squared                0.781       MSE              33957.712
## Adj. R-Squared           0.758       Coef. Var           27.839
## Pred R-Squared           0.700       AIC                730.620
## MAE                    137.656       SBC                744.543
## ------------------------------------------------------------------
##  RMSE: Root Mean Square Error
##  MSE: Mean Square Error
##  MAE: Mean Absolute Error
##  AIC: Akaike Information Criteria
##  SBC: Schwarz Bayesian Criteria
##
##                                  ANOVA
## ----------------------------------------------------------------------
##                  Sum of
##                 Squares        DF    Mean Square      F         Sig.
## ----------------------------------------------------------------------
## Regression    6535804.090       5    1307160.818    34.217    0.0000
## Residual      1833716.447      48      38202.426
## Total         8369520.537      53
## ----------------------------------------------------------------------
##
##                                       Parameter Estimates
## -------------------------------------------------------------------------------------------------
##       model         Beta    Std. Error    Std. Beta      t        Sig        lower       upper
## -------------------------------------------------------------------------------------------------
## (Intercept)    -1178.330       208.682                 -5.647    0.000    -1597.914    -758.746
##  liver_test       58.064        40.144        0.156     1.446    0.155      -22.652     138.779
##   alc_heavy      317.848        71.634        0.314     4.437    0.000      173.818     461.878
## enzyme_test        9.748         1.656        0.521     5.887    0.000        6.419      13.077
##      pindex        8.924         1.808        0.380     4.935    0.000        5.288      12.559
##         bcs       59.864        23.060        0.241     2.596    0.012       13.498     106.230
## -------------------------------------------------------------------------------------------------
```

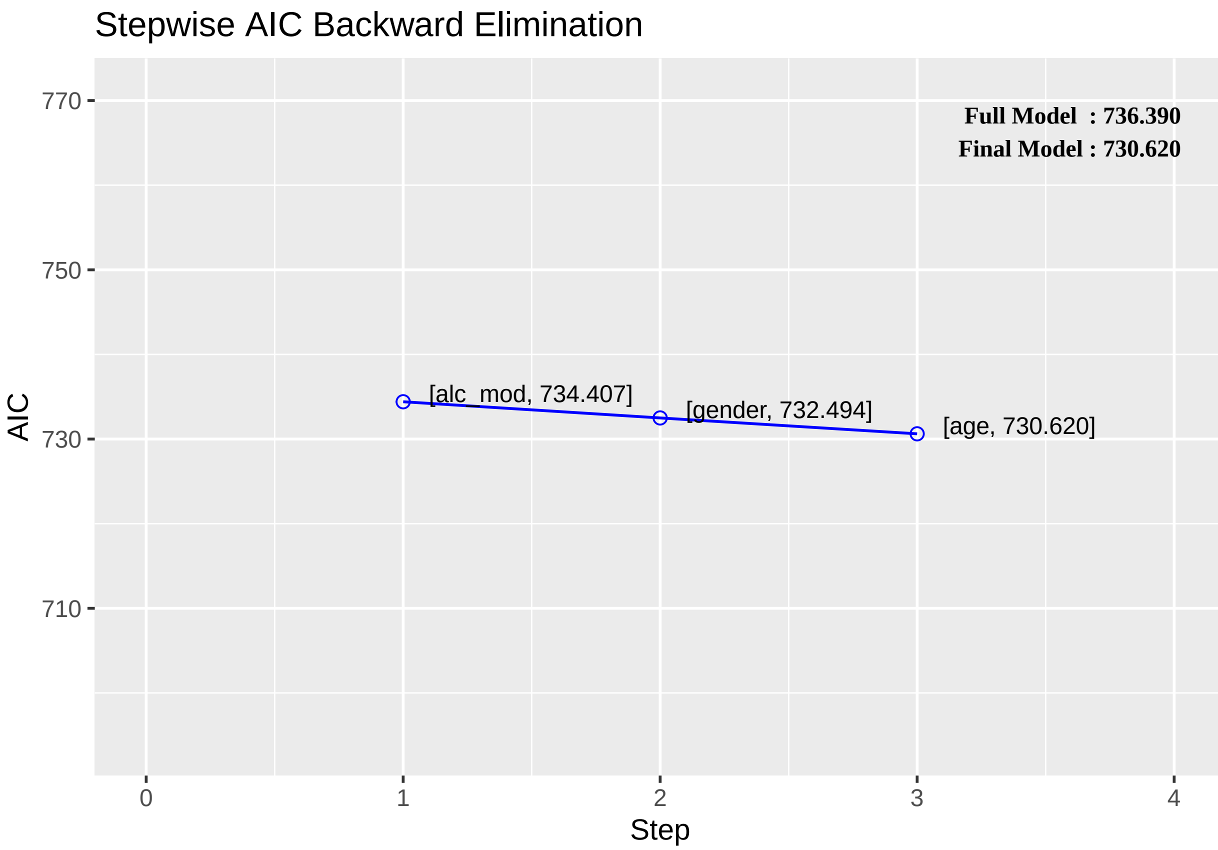
Stepwise AIC Backward Regression

Build regression model from a set of candidate predictor variables by removing predictors based on Akaike Information Criteria, in a stepwise manner until there is no variable left to remove any more. The model should include all the candidate predictor variables. If details is set to TRUE, each step is displayed.

```r
k7 <- ols_step_backward_aic(exp2)
k7
```

```
##
##
##                              Stepwise Summary
## ------------------------------------------------------------------------
## Step    Variable        AIC        SBC       SBIC        R2       Adj. R2
## ------------------------------------------------------------------------
##  0      Full Model    736.390    756.280    586.665    0.78184    0.74305
##  1      alc_mod       734.407    752.308    583.884    0.78177    0.74856
##  2      gender        732.494    748.406    581.290    0.78142    0.75351
##  3      age           730.620    744.543    578.844    0.78091    0.75808
## ------------------------------------------------------------------------
##
## Final Model Output
## ------------------
##
##                            Model Summary
## ------------------------------------------------------------------
## R                           0.884       RMSE                184.276
## R-Squared                   0.781       MSE               33957.712
## Adj. R-Squared              0.758       Coef. Var            27.839
## Pred R-Squared              0.700       AIC                 730.620
## MAE                       137.656       SBC                 744.543
## ------------------------------------------------------------------
##  RMSE: Root Mean Square Error
##  MSE: Mean Square Error
##  MAE: Mean Absolute Error
##  AIC: Akaike Information Criteria
##  SBC: Schwarz Bayesian Criteria
##
##                                  ANOVA
## ----------------------------------------------------------------------
##                    Sum of
##                   Squares        DF    Mean Square      F         Sig.
## ----------------------------------------------------------------------
## Regression    6535804.090         5    1307160.818    34.217    0.0000
## Residual      1833716.447        48      38202.426
## Total         8369520.537        53
## ----------------------------------------------------------------------
##
##                                       Parameter Estimates
## ----------------------------------------------------------------------------------------------
##       model         Beta    Std. Error    Std. Beta      t        Sig        lower       upper
## ----------------------------------------------------------------------------------------------
## (Intercept)    -1178.330       208.682                 -5.647    0.000    -1597.914    -758.746
##         bcs       59.864        23.060        0.241     2.596    0.012       13.498     106.230
##      pindex        8.924         1.808        0.380     4.935    0.000        5.288      12.559
## enzyme_test        9.748         1.656        0.521     5.887    0.000        6.419      13.077
##  liver_test       58.064        40.144        0.156     1.446    0.155      -22.652     138.779
##   alc_heavy      317.848        71.634        0.314     4.437    0.000      173.818     461.878
## ----------------------------------------------------------------------------------------------
```

```r
plot(k7)
```




```r
ols_step_backward_aic(exp2, details = TRUE)
```

```
## Backward Elimination Method
## ---------------------------
##
## Candidate Terms:
##
## 1. bcs
## 2. pindex
## 3. enzyme_test
## 4. liver_test
## 5. age
## 6. gender
## 7. alc_mod
## 8. alc_heavy
##
##
## Step     => 0
## Model    => y ~ bcs + pindex + enzyme_test + liver_test + age + gender + alc_mod + alc_heavy
## AIC      => 736.3899
##
## Initiating stepwise selection...
##
##                    Table: Removing Existing Variables
## -------------------------------------------------------------------------
## Predictor      DF      AIC        SBC       SBIC        R2        Adj. R2
## -------------------------------------------------------------------------
```

```
## alc_mod         1    734.407    752.308    584.276    0.78177    0.74856
## gender          1    734.478    752.379    584.323    0.78148    0.74823
## age             1    734.544    752.445    584.367    0.78121    0.74792
## liver_test      1    735.878    753.779    585.255    0.77574    0.74162
## bcs             1    741.677    759.577    589.203    0.75032    0.71233
## alc_heavy       1    749.210    767.111    594.541    0.71294    0.66926
## pindex          1    756.624    774.525    600.014    0.67070    0.62059
## enzyme_test     1    763.557    781.458    605.318    0.62559    0.56861
## ------------------------------------------------------------------------
##
## Step     => 1
## Removed  => alc_mod
## Model    => y ~ bcs + pindex + enzyme_test + liver_test + age + gender + alc_heavy
## AIC      => 734.4068
##
##                    Table: Removing Existing Variables
## ------------------------------------------------------------------------
## Predictor      DF      AIC        SBC       SBIC        R2       Adj. R2
## ------------------------------------------------------------------------
## gender          1    732.494    748.406    581.938    0.78142    0.75351
## age             1    732.551    748.463    581.978    0.78119    0.75325
## liver_test      1    733.921    749.833    582.951    0.77556    0.74691
## bcs             1    739.677    755.589    587.106    0.75032    0.71845
## alc_heavy       1    750.486    766.398    595.217    0.69499    0.65605
## pindex          1    754.759    770.671    598.530    0.66987    0.62773
## enzyme_test     1    761.595    777.507    603.950    0.62532    0.57749
## ------------------------------------------------------------------------
##
## Step     => 2
## Removed  => gender
## Model    => y ~ bcs + pindex + enzyme_test + liver_test + age + alc_heavy
## AIC      => 732.4942
##
##                    Table: Removing Existing Variables
## ------------------------------------------------------------------------
## Predictor      DF      AIC        SBC       SBIC        R2       Adj. R2
## ------------------------------------------------------------------------
## age             1    730.620    744.543    579.638    0.78091    0.75808
## liver_test      1    732.339    746.262    580.934    0.77382    0.75026
## bcs             1    737.680    751.603    585.012    0.75030    0.72429
## alc_heavy       1    748.486    762.409    593.500    0.69499    0.66322
## pindex          1    752.777    766.700    596.959    0.66976    0.63536
## enzyme_test     1    759.596    773.518    602.553    0.62532    0.58629
## ------------------------------------------------------------------------
##
## Step     => 3
## Removed  => age
## Model    => y ~ bcs + pindex + enzyme_test + liver_test + alc_heavy
## AIC      => 730.6204
##
##                    Table: Removing Existing Variables
## ------------------------------------------------------------------------
## Predictor      DF      AIC        SBC       SBIC        R2       Adj. R2
## ------------------------------------------------------------------------
```

```
## liver_test      1    730.924    742.858    579.087    0.77136    0.75269
## bcs             1    735.715    747.649    582.943    0.75015    0.72975
## alc_heavy       1    747.181    759.114    592.362    0.69104    0.66582
## pindex          1    750.782    762.716    595.377    0.66973    0.64277
## enzyme_test     1    757.971    769.905    601.477    0.62270    0.59190
## ------------------------------------------------------------------------
##
##
## No more variables to be removed.
##
## Variables Removed:
##
## => alc_mod
## => gender
## => age

##
##
##                              Stepwise Summary
## -------------------------------------------------------------------------
## Step    Variable        AIC        SBC       SBIC        R2       Adj. R2
## -------------------------------------------------------------------------
##  0      Full Model    736.390    756.280    586.665    0.78184    0.74305
##  1      alc_mod       734.407    752.308    583.884    0.78177    0.74856
##  2      gender        732.494    748.406    581.290    0.78142    0.75351
##  3      age           730.620    744.543    578.844    0.78091    0.75808
## -------------------------------------------------------------------------
##
## Final Model Output
## ------------------
##
##                            Model Summary
## -------------------------------------------------------------------
## R                         0.884       RMSE                 184.276
## R-Squared                 0.781       MSE                33957.712
## Adj. R-Squared            0.758       Coef. Var             27.839
## Pred R-Squared            0.700       AIC                  730.620
## MAE                     137.656       SBC                  744.543
## -------------------------------------------------------------------
##  RMSE: Root Mean Square Error
##  MSE: Mean Square Error
##  MAE: Mean Absolute Error
##  AIC: Akaike Information Criteria
##  SBC: Schwarz Bayesian Criteria
##
##                                  ANOVA
## -----------------------------------------------------------------------
##                   Sum of
##                  Squares        DF    Mean Square      F          Sig.
## -----------------------------------------------------------------------
## Regression    6535804.090        5    1307160.818    34.217    0.0000
## Residual      1833716.447       48      38202.426
## Total         8369520.537       53
## -----------------------------------------------------------------------
```

```
##
##                                      Parameter Estimates
## -----------------------------------------------------------------------------------------------
##       model         Beta    Std. Error    Std. Beta      t        Sig         lower        upper
## -----------------------------------------------------------------------------------------------
## (Intercept)    -1178.330       208.682                -5.647    0.000     -1597.914     -758.746
##         bcs       59.864        23.060        0.241    2.596    0.012        13.498      106.230
##      pindex        8.924         1.808        0.380    4.935    0.000         5.288       12.559
## enzyme_test        9.748         1.656        0.521    5.887    0.000         6.419       13.077
##  liver_test       58.064        40.144        0.156    1.446    0.155       -22.652      138.779
##   alc_heavy      317.848        71.634        0.314    4.437    0.000       173.818      461.878
## -----------------------------------------------------------------------------------------------
```

Stepwise AIC Regression

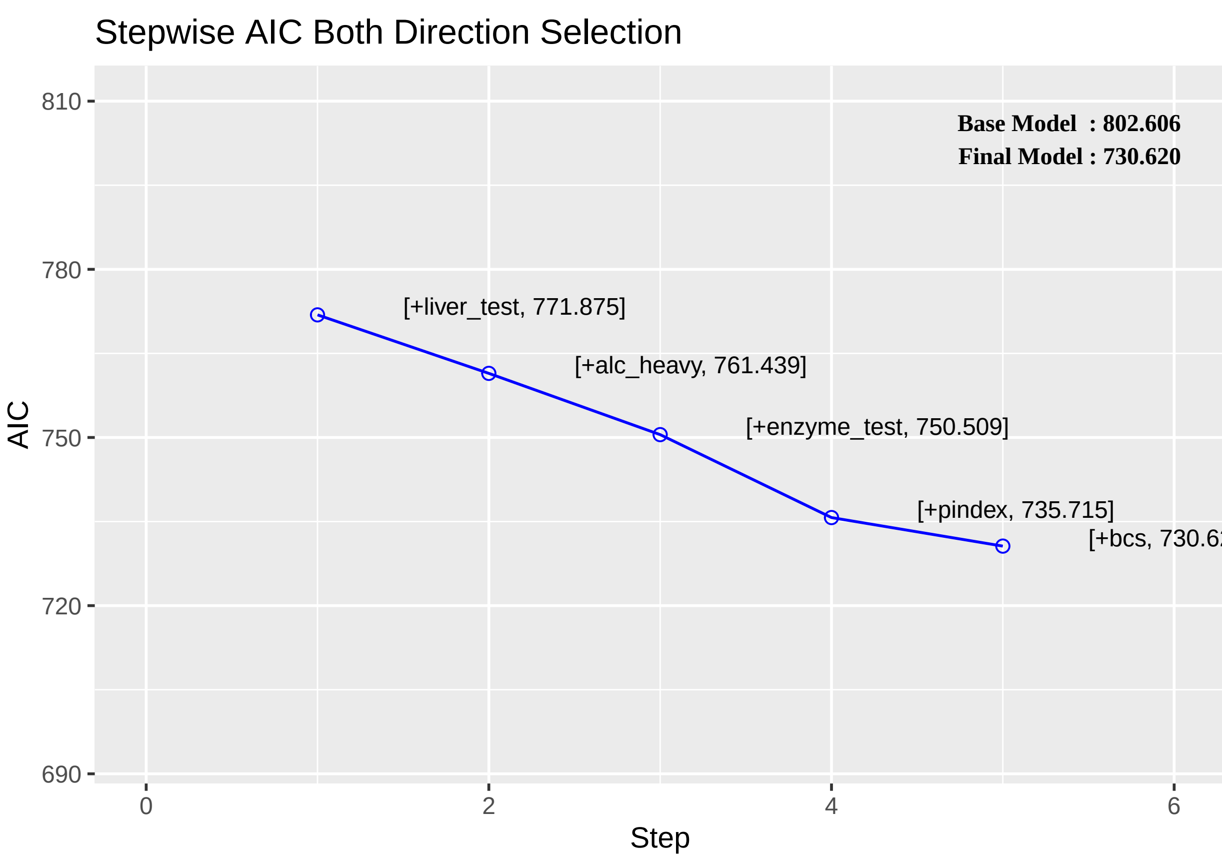
Build regression model from a set of candidate predictor variables by entering and removing predictors based on Akaike Information Criteria, in a stepwise manner until there is no variable left to enter or remove any more. The model should include all the candidate predictor variables. If details is set to TRUE, each step is displayed.

```r
k8<-ols_step_both_aic(exp2)
k8
```

```
##
##
##                                 Stepwise Summary
## -----------------------------------------------------------------------------
## Step    Variable            AIC        SBC       SBIC        R2       Adj. R2
## -----------------------------------------------------------------------------
##  0      Base Model        802.606    806.584    646.794    0.00000    0.00000
##  1      liver_test (+)    771.875    777.842    616.009    0.45454    0.44405
##  2      alc_heavy (+)     761.439    769.395    605.506    0.56674    0.54975
##  3      enzyme_test (+)   750.509    760.454    595.297    0.65900    0.63854
##  4      pindex (+)        735.715    747.649    582.943    0.75015    0.72975
##  5      bcs (+)           730.620    744.543    579.638    0.78091    0.75808
## -----------------------------------------------------------------------------
##
## Final Model Output
## ------------------
##
##                            Model Summary
## ------------------------------------------------------------------
## R                        0.884       RMSE               184.276
## R-Squared                0.781       MSE              33957.712
## Adj. R-Squared           0.758       Coef. Var           27.839
## Pred R-Squared           0.700       AIC                730.620
## MAE                    137.656       SBC                744.543
## ------------------------------------------------------------------
##  RMSE: Root Mean Square Error
##  MSE: Mean Square Error
##  MAE: Mean Absolute Error
##  AIC: Akaike Information Criteria
##  SBC: Schwarz Bayesian Criteria
##
##                                  ANOVA
```

```
## --------------------------------------------------------------------
##                    Sum of
##                   Squares        DF    Mean Square      F         Sig.
## --------------------------------------------------------------------
## Regression    6535804.090        5    1307160.818    34.217    0.0000
## Residual      1833716.447       48      38202.426
## Total         8369520.537       53
## --------------------------------------------------------------------
##
##                                   Parameter Estimates
## ----------------------------------------------------------------------------------------------
##       model         Beta    Std. Error    Std. Beta       t        Sig        lower       upper
## ----------------------------------------------------------------------------------------------
## (Intercept)    -1178.330       208.682                 -5.647    0.000    -1597.914    -758.746
##  liver_test       58.064        40.144        0.156     1.446    0.155      -22.652     138.779
##   alc_heavy      317.848        71.634        0.314     4.437    0.000      173.818     461.878
## enzyme_test        9.748         1.656        0.521     5.887    0.000        6.419      13.077
##      pindex        8.924         1.808        0.380     4.935    0.000        5.288      12.559
##         bcs       59.864        23.060        0.241     2.596    0.012       13.498     106.230
## ----------------------------------------------------------------------------------------------
```

```r
plot(k8)
```




```r
ols_step_both_aic(exp2, details = TRUE)
```

```
## Stepwise Selection Method
## -------------------------
##
```

```
## Candidate Terms:
##
## 1. bcs
## 2. pindex
## 3. enzyme_test
## 4. liver_test
## 5. age
## 6. gender
## 7. alc_mod
## 8. alc_heavy
##
##
## Step     => 0
## Model    => y ~ 1
## AIC      => 802.606
##
## Initiating stepwise selection...
##
##                        Table: Adding New Variables
## -------------------------------------------------------------------------
## Predictor      DF      AIC        SBC       SBIC        R2       Adj. R2
## -------------------------------------------------------------------------
## bcs             1    797.697    803.664    640.655    0.12010     0.10318
## pindex          1    794.100    800.067    637.196    0.17680     0.16097
## enzyme_test     1    782.629    788.596    626.220    0.33435     0.32154
## liver_test      1    771.875    777.842    616.009    0.45454     0.44405
## age             1    803.834    809.801    646.572    0.01420    -0.00476
## gender          1    802.956    808.923    645.725    0.03009     0.01143
## alc_mod         1    802.828    808.795    645.601    0.03239     0.01378
## alc_heavy       1    794.301    800.268    637.389    0.17373     0.15784
## -------------------------------------------------------------------------
##
## Step     => 1
## Added    => liver_test
## Model    => y ~ liver_test
## AIC      => 771.8753
##
##                       Table: Adding New Variables
## ------------------------------------------------------------------------
## Predictor      DF      AIC        SBC       SBIC        R2       Adj. R2
## ------------------------------------------------------------------------
## bcs             1    773.867    781.823    616.961    0.45462    0.43323
## pindex          1    770.387    778.343    613.737    0.48866    0.46861
## enzyme_test     1    762.077    770.033    606.090    0.56159    0.54440
## age             1    773.831    781.787    616.928    0.45498    0.43361
## gender          1    773.802    781.758    616.901    0.45528    0.43391
## alc_mod         1    771.141    779.097    614.435    0.48147    0.46113
## alc_heavy       1    761.439    769.395    605.506    0.56674    0.54975
## ------------------------------------------------------------------------
##
## Step     => 2
## Added    => alc_heavy
## Model    => y ~ liver_test + alc_heavy
## AIC      => 761.4394
```

```
##
##                    Table: Removing Existing Variables
## ---------------------------------------------------------------------------
## Predictor     DF      AIC        SBC       SBIC        R2       Adj. R2
## ---------------------------------------------------------------------------
## liver_test     1    794.301    800.268    637.389    0.17373    0.15784
## alc_heavy      1    771.875    777.842    616.009    0.45454    0.44405
## ---------------------------------------------------------------------------
##
##                        Table: Adding New Variables
## ----------------------------------------------------------------------------
## Predictor      DF      AIC        SBC       SBIC        R2       Adj. R2
## ----------------------------------------------------------------------------
## bcs             1    763.063    773.008    606.379    0.56975    0.54394
## pindex          1    756.125    766.070    600.225    0.62163    0.59892
## enzyme_test     1    750.509    760.454    595.297    0.65900    0.63854
## age             1    763.110    773.055    606.421    0.56938    0.54354
## gender          1    763.433    773.378    606.709    0.56679    0.54080
## alc_mod         1    763.428    773.373    606.704    0.56683    0.54084
## ----------------------------------------------------------------------------
##
## Step     => 3
## Added    => enzyme_test
## Model    => y ~ liver_test + alc_heavy + enzyme_test
## AIC      => 750.5089
##
##                     Table: Removing Existing Variables
## ----------------------------------------------------------------------------
## Predictor      DF      AIC        SBC       SBIC        R2       Adj. R2
## ----------------------------------------------------------------------------
## liver_test      1    773.555    781.511    616.671    0.45777    0.43650
## alc_heavy       1    762.077    770.033    606.090    0.56159    0.54440
## enzyme_test     1    761.439    769.395    605.506    0.56674    0.54975
## ----------------------------------------------------------------------------
##
##                       Table: Adding New Variables
## --------------------------------------------------------------------------
## Predictor    DF      AIC        SBC       SBIC        R2       Adj. R2
## --------------------------------------------------------------------------
## bcs           1    750.782    762.716    595.377    0.66973    0.64277
## pindex        1    735.715    747.649    582.943    0.75015    0.72975
## age           1    752.416    764.350    596.755    0.65959    0.63180
## gender        1    752.509    764.443    596.833    0.65900    0.63116
## alc_mod       1    752.403    764.337    596.743    0.65967    0.63189
## --------------------------------------------------------------------------
##
## Step     => 4
## Added    => pindex
## Model    => y ~ liver_test + alc_heavy + enzyme_test + pindex
## AIC      => 735.7146
##
##                     Table: Removing Existing Variables
## ----------------------------------------------------------------------------
## Predictor      DF      AIC        SBC       SBIC        R2       Adj. R2
```

```
## ------------------------------------------------------------------------
## liver_test      1    748.167    758.112    593.257    0.67347    0.65388
## alc_heavy       1    755.099    765.044    599.321    0.62875    0.60647
## enzyme_test     1    756.125    766.070    600.225    0.62163    0.59892
## pindex          1    750.509    760.454    595.297    0.65900    0.63854
## ------------------------------------------------------------------------
##
##                       Table: Adding New Variables
## ----------------------------------------------------------------------
## Predictor    DF      AIC        SBC       SBIC        R2       Adj. R2
## ----------------------------------------------------------------------
## bcs           1    730.620    744.543    579.638    0.78091    0.75808
## age           1    737.680    751.603    585.012    0.75030    0.72429
## gender        1    737.712    751.635    585.036    0.75016    0.72413
## alc_mod       1    737.713    751.636    585.037    0.75015    0.72413
## ----------------------------------------------------------------------
##
## Step     => 5
## Added    => bcs
## Model    => y ~ liver_test + alc_heavy + enzyme_test + pindex + bcs
## AIC      => 730.6204
##
##                     Table: Removing Existing Variables
## ------------------------------------------------------------------------
## Predictor      DF      AIC        SBC       SBIC        R2       Adj. R2
## ------------------------------------------------------------------------
## liver_test      1    730.924    742.858    579.087    0.77136    0.75269
## alc_heavy       1    747.181    759.114    592.362    0.69104    0.66582
## enzyme_test     1    757.971    769.905    601.477    0.62270    0.59190
## pindex          1    750.782    762.716    595.377    0.66973    0.64277
## bcs             1    735.715    747.649    582.943    0.75015    0.72975
## ------------------------------------------------------------------------
##
##                       Table: Adding New Variables
## ----------------------------------------------------------------------
## Predictor    DF      AIC        SBC       SBIC        R2       Adj. R2
## ----------------------------------------------------------------------
## age           1    732.494    748.406    581.938    0.78142    0.75351
## gender        1    732.551    748.463    581.978    0.78119    0.75325
## alc_mod       1    732.614    748.526    582.023    0.78093    0.75297
## ----------------------------------------------------------------------
##
##
## No more variables to be added or removed.
##
## Variables Selected:
##
## => liver_test
## => alc_heavy
## => enzyme_test
## => pindex
## => bcs

##
```

```
##
##                                 Stepwise Summary
## --------------------------------------------------------------------------------
## Step    Variable             AIC        SBC       SBIC        R2       Adj. R2
## --------------------------------------------------------------------------------
##  0      Base Model         802.606    806.584    646.794    0.00000    0.00000
##  1      liver_test (+)     771.875    777.842    616.009    0.45454    0.44405
##  2      alc_heavy (+)      761.439    769.395    605.506    0.56674    0.54975
##  3      enzyme_test (+)    750.509    760.454    595.297    0.65900    0.63854
##  4      pindex (+)         735.715    747.649    582.943    0.75015    0.72975
##  5      bcs (+)            730.620    744.543    579.638    0.78091    0.75808
## --------------------------------------------------------------------------------
##
## Final Model Output
## ------------------
##
##                            Model Summary
## -------------------------------------------------------------------
## R                          0.884       RMSE                184.276
## R-Squared                  0.781       MSE               33957.712
## Adj. R-Squared             0.758       Coef. Var            27.839
## Pred R-Squared             0.700       AIC                 730.620
## MAE                      137.656       SBC                 744.543
## -------------------------------------------------------------------
##  RMSE: Root Mean Square Error
##  MSE: Mean Square Error
##  MAE: Mean Absolute Error
##  AIC: Akaike Information Criteria
##  SBC: Schwarz Bayesian Criteria
##
##                                  ANOVA
## ----------------------------------------------------------------------
##                  Sum of
##                 Squares        DF    Mean Square      F         Sig.
## ----------------------------------------------------------------------
## Regression    6535804.090       5    1307160.818    34.217    0.0000
## Residual      1833716.447      48      38202.426
## Total         8369520.537      53
## ----------------------------------------------------------------------
##
##                                       Parameter Estimates
## ----------------------------------------------------------------------------------------------
##       model         Beta    Std. Error    Std. Beta      t        Sig         lower       upper
## ----------------------------------------------------------------------------------------------
## (Intercept)    -1178.330       208.682                 -5.647    0.000    -1597.914    -758.746
##  liver_test       58.064        40.144        0.156     1.446    0.155      -22.652     138.779
##   alc_heavy      317.848        71.634        0.314     4.437    0.000      173.818     461.878
## enzyme_test        9.748         1.656        0.521     5.887    0.000        6.419      13.077
##      pindex        8.924         1.808        0.380     4.935    0.000        5.288      12.559
##         bcs       59.864        23.060        0.241     2.596    0.012       13.498     106.230
## ----------------------------------------------------------------------------------------------
```

##Example of using leaps library

We illustrate backward elimination on some data on the 50 states from the 1970s. The data were collected from U.S. Bureau of the Census. We will take life expectancy as the response and the remaining variables as predictors:

```
data(state)
state.x77[1:10,]
```

```
##              Population Income Illiteracy Life Exp Murder HS Grad Frost   Area
## Alabama            3615   3624        2.1    69.05   15.1    41.3    20  50708
## Alaska              365   6315        1.5    69.31   11.3    66.7   152 566432
## Arizona            2212   4530        1.8    70.55    7.8    58.1    15 113417
## Arkansas           2110   3378        1.9    70.66   10.1    39.9    65  51945
## California        21198   5114        1.1    71.71   10.3    62.6    20 156361
## Colorado           2541   4884        0.7    72.06    6.8    63.9   166 103766
## Connecticut        3100   5348        1.1    72.48    3.1    56.0   139   4862
## Delaware            579   4809        0.9    70.06    6.2    54.6   103   1982
## Florida            8277   4815        1.3    70.66   10.7    52.6    11  54090
## Georgia            4931   4091        2.0    68.54   13.9    40.6    60  58073
```

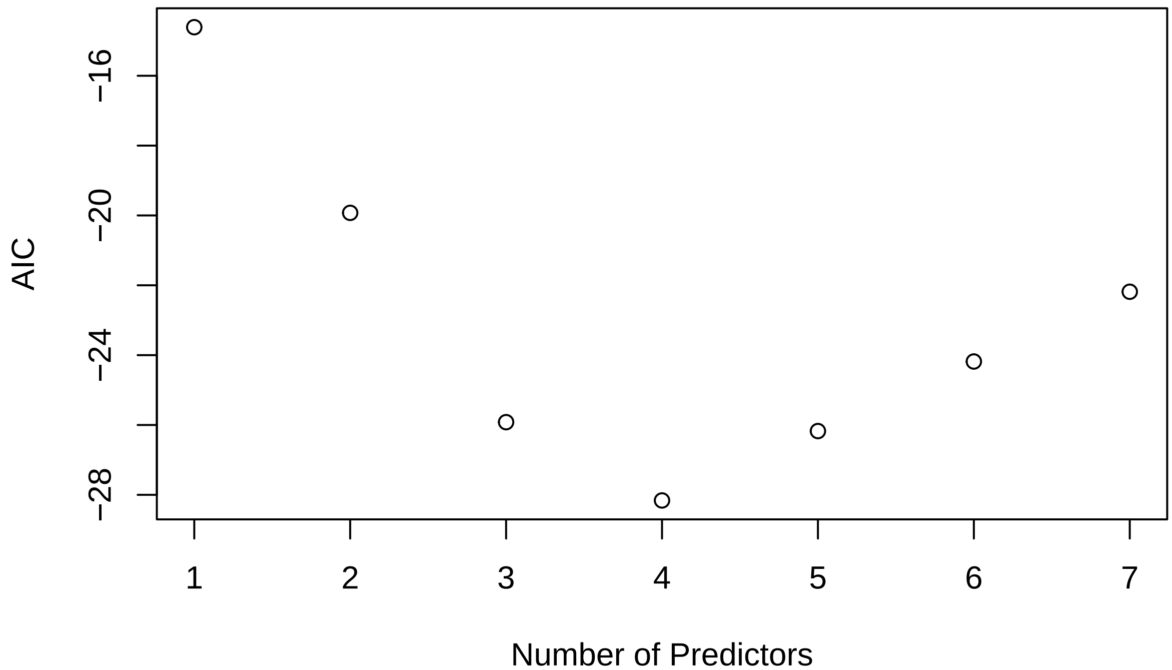
```
state<-data.frame(state.x77)
require(leaps)
```

```
## Loading required package: leaps
```

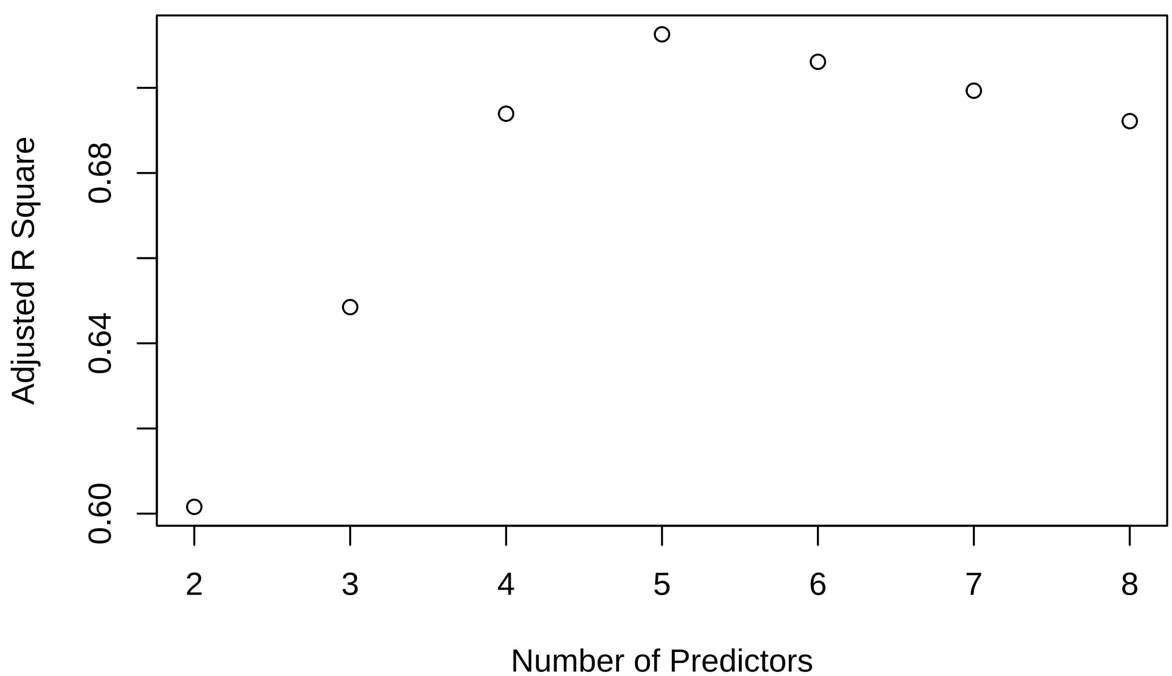
```
b<-regsubsets(Life.Exp~.,data=state)
rs<-summary(b)
rs$which
```

```
##   (Intercept) Population Income Illiteracy Murder HS.Grad Frost  Area
## 1        TRUE      FALSE  FALSE      FALSE   TRUE   FALSE FALSE FALSE
## 2        TRUE      FALSE  FALSE      FALSE   TRUE    TRUE FALSE FALSE
## 3        TRUE      FALSE  FALSE      FALSE   TRUE    TRUE  TRUE FALSE
## 4        TRUE       TRUE  FALSE      FALSE   TRUE    TRUE  TRUE FALSE
## 5        TRUE       TRUE   TRUE      FALSE   TRUE    TRUE  TRUE FALSE
## 6        TRUE       TRUE   TRUE       TRUE   TRUE    TRUE  TRUE FALSE
## 7        TRUE       TRUE   TRUE       TRUE   TRUE    TRUE  TRUE  TRUE
```

```
AIC<-50*log(rs$rss/50)+(2:8)*2
plot(AIC~I(1:7),ylab="AIC",xlab="Number of Predictors")
```



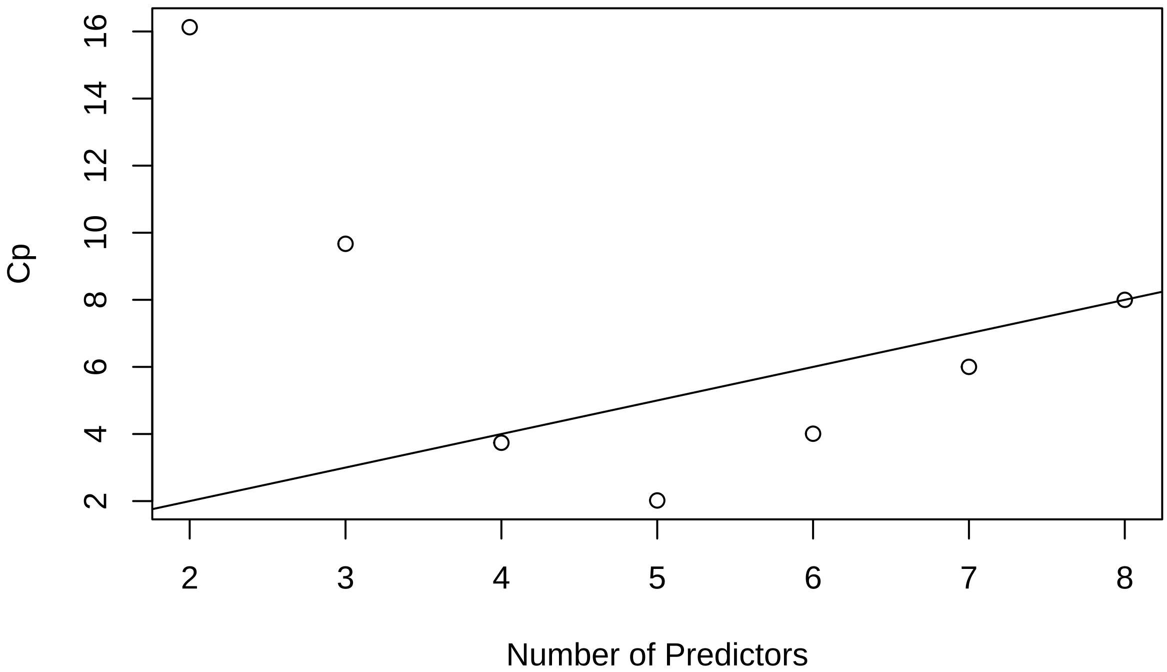

```
plot(2:8,rs$adjr2,xlab="Number of Predictors",ylab="Adjusted R Square")
```




```
which.max(rs$adjr2)
```

```
## [1] 4
```

```
plot(2:8,rs$cp,xlab="Number of Predictors",ylab="Cp")
abline(0,1)
```

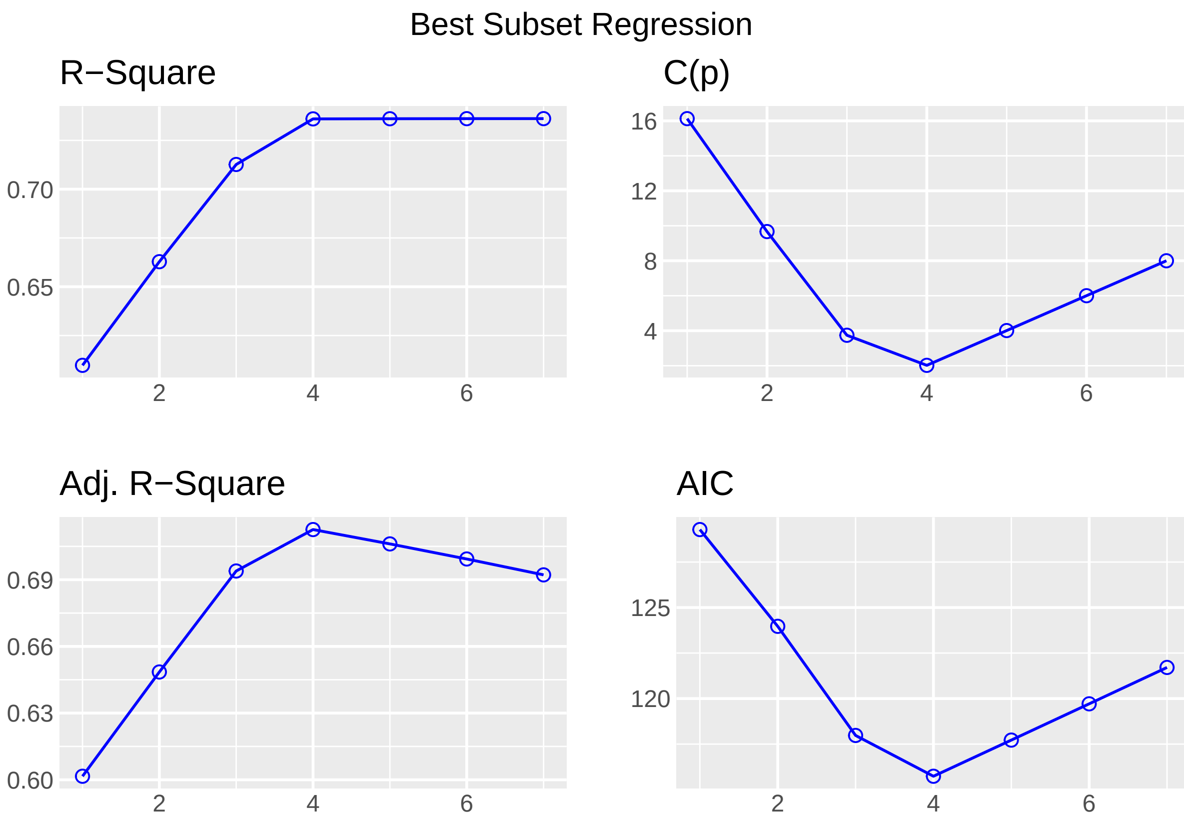


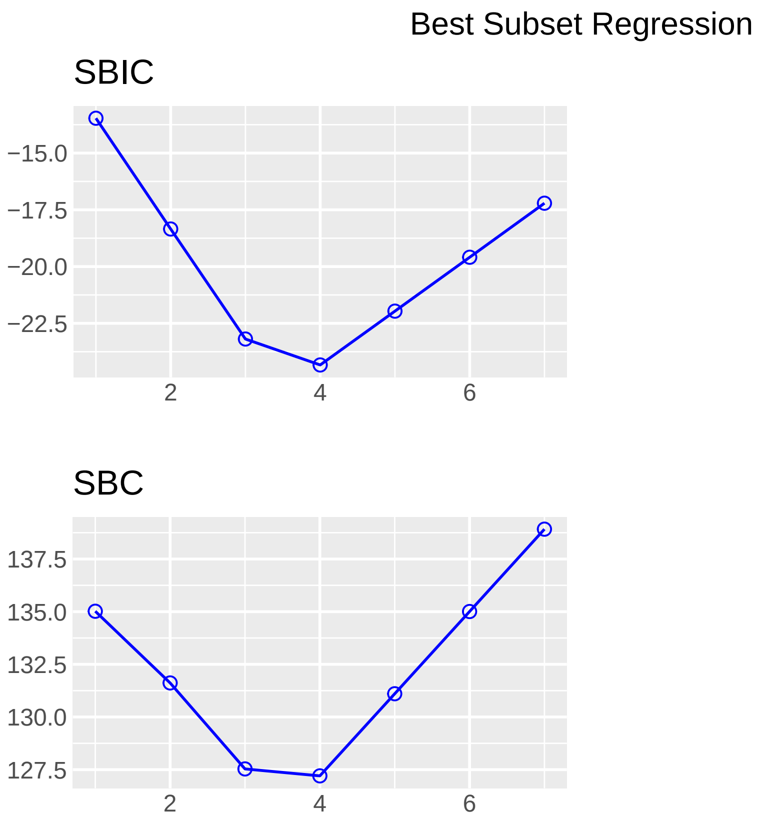

```r
st1<-lm(Life.Exp~.,data=state)
res1<-ols_step_best_subset(st1)
res1
```

```
##                          Best Subsets Regression
## ----------------------------------------------------------------------
## Model Index    Predictors
## ----------------------------------------------------------------------
##      1         Murder
##      2         Murder HS.Grad
##      3         Murder HS.Grad Frost
##      4         Population Murder HS.Grad Frost
##      5         Population Income Murder HS.Grad Frost
##      6         Population Income Illiteracy Murder HS.Grad Frost
##      7         Population Income Illiteracy Murder HS.Grad Frost Area
## ----------------------------------------------------------------------
##
##                                                    Subsets Regression Summary
## ------------------------------------------------------------------------------------------------------------------------------
##                        Adj.        Pred
## Model    R-Square    R-Square    R-Square     C(p)        AIC         SBIC         SBC         MSEP
## ------------------------------------------------------------------------------------------------------------------------------
##   1        0.6097      0.6016      0.5789    16.1268    129.2846    -13.4662    135.0207    35.8984
##   2        0.6628      0.6485      0.6071     9.6699    123.9684    -18.3451    131.6165    31.6860
##   3        0.7127      0.6939      0.6467     3.7399    117.9743    -23.1918    127.5344    27.6043
##   4        0.7360      0.7126      0.6642     2.0197    115.7326    -24.3342    127.2048    25.9355
##   5        0.7361      0.7061      0.6453     4.0087    117.7196    -21.9639    131.1038    26.5317
##   6        0.7361      0.6993      0.6268     6.0020    119.7116    -19.5889    135.0077    27.1591
##   7        0.7362      0.6922       0.532     8.0000    121.7092    -17.2096    138.9174    27.8202
## ------------------------------------------------------------------------------------------------------------------------------
## AIC: Akaike Information Criteria
##  SBIC: Sawa's Bayesian Information Criteria
##  SBC: Schwarz Bayesian Criteria
```

```
##   MSEP: Estimated error of prediction, assuming multivariate normality
##   FPE: Final Prediction Error
##   HSP: Hocking's Sp
##   APC: Amemiya Prediction Criteria
```

```r
plot(res1)
```





The competition is between the four-parameter, three-predictor, model including frost, high school graduation and murder and the model also including population.

Both models are on or below the Cp = p line, indicating good fits. The choice is between the smaller model and the larger model, which fits a little better.

Some even larger models fit in the sense that they are on or below the Cp = p line, but we would not opt for these in the presence of smaller models that fit.